AIA · AGENTIC AI ANALYTICS · METHODOLOGY

The business-decision study
Study type 2: a cost model of options

How AIA takes a client's decision from *Zadání* to a report in which every number carries its source and its grade. Version 3, written bottom up from the ChemVitro exemplar.

THE SPINE · EIGHT STEPS

Zadání · *Data* · *Rešerše* · *Předpoklady* · *Model* · *Výsledky* · *Sociomapa* · *Osnova a Zpráva*

The navigator shows nine screens; the outline and the report are one step here. The client's rule is written at the first step and applied, as written, at the sixth. Every number in the report carries its source and one of seven grades. A method the register does not hold stays provisional.

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Replaces version 2 (36 pages, 5 November 2026), which recorded the methodology review R1–R10
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Case pack: `prototypes/case-chemvitro/` (branch `chore/case-chemvitro`) · Design canvas: `prototypes/obchodni-model/`
Agrees with the MVP scope: `prototypes/obchodni-model/MVP-SCOPE.md` (session A, branch `chore/mvp-r2-a`, 8 November 2026)

This is design, not product code. Section 10 says what exists in AIA today and what does not. Every client, site, person and number in the exemplar is fictional.

CONTENTS

What is in this document

What this is 3

1 The spine 4

1.1 The shape of each step chapter 4

1.2 The scope's acceptance and waves 5

1.3 The rules the whole spine keeps 5

2 Zadání: the decision and its rule 6

3 Data: the client's files under contract 8

4 Rešerše: public inputs with a state 10

5 Předpoklady: one table of inputs 12

6 Model: one computation for every option 14

7 Výsledky: the rule applied, and what could change it 16

8 Sociomapa: the futures map 19

9 Osnova a Zpráva: the outline, then the report 21

10 Built vs designed 24

10.1 The state on 8 November 2026 24

10.2 The plans that carry the build 24

10.3 Three honest statements 24

11 Perspectives, components and the two maps 25

11.1 Perspectives and components 25

11.2 The futures map 26

11.3 The weighting-scenario map (Mapa scénářů vah, board MV9) 26

12 Grades, sources and the release gate 28

12.1 The exemplar's seven grades 28

12.2 The four attributes behind a grade 28

12.3 Sources, numbered per chapter 28

12.4 The provisional state 29

12.5 The release gate (R9, proposed) 29

13 Governance 30

13.1 The validity review (R1) 30

13.2 Register entries the exemplar waits on 30

13.3 The methodologist's open questions 31

13.4 The owner's open decisions 31

13.5 Change propagation (K4) 32

A Glossary 33

B The exemplar in numbers 34

B.1 T1 · The rule at 8 % 34

B.2 T4 · Finance, base case 34

B.3 T6 · Operating cost per tonne in 2032, base case, EUR/t 34

B.4 T7 · Relations between sites across the futures 34

B.5 T8 · Shared failure of the two leading sites 35

B.6 The decision boundary and the input table in brief 35

C Board index 36

BEFORE THE CHAPTERS

What this is

On 8 November 2026 the executive chose report 2 of the ChemVitro case study as the report AIA must produce. This document is the methodology behind that report, step by step, and the measure the MVP will be held to.

The three outcomes of the pivot

Outcome	Where it lives	Who
One MVP prototype	The Design canvas *AIA Obchodní model*, `prototypes/obchodni-model/`; thirty-one boards (MV9 to come), Czech, one path from *Zadání* to *Zpráva*; since 8 November the boards carry the ChemVitro case beside the Velis case, each section marked with its case	sessions A–E
One methodology PDF	This document, `prototypes/obchodni-model/methodology/`	session F
One MVP scope	`prototypes/obchodni-model/MVP-SCOPE.md` and board MV0; the methodology and the scope must agree	session A

What the MVP produces

Study type 2, *Model nákladů lokality*: a decision between options, judged by a rule the client wrote before any result, on a cost model computed by code, with the uncertainty of every input carried through to the verdict. The exemplar is a type-2 report on a fictional client, ChemVitro, choosing among eight sites for a chemical plant or keeping its supplier. Its parts carry the ids R2-P01 to R2-P18 and its tables T1 to T9; this document cites them by those ids.

Types 1 and 3 of the case pack (the situation report without a model; the model beside a worker survey) are not the MVP. The survey lane is frozen (ADR 0046, decision 2).

How to read it

- Chapter 1 is the spine: the eight steps in one table (the canvas's navigator shows nine; the outline and the report are one step).
- Chapters 2 to 9 take one step each, all in the same shape: what goes in, what comes out, the checks, how the exemplar did it, what AIA has today, what is new, and what the ChemVitro acceptance expects.
- Chapter 10 says, truthfully, what is built and what is designed. No type-2 code is on `develop`.
- Chapters 11 to 13 cross the steps: the business perspectives and the two Sociomaps; grades, sources and the release gate; governance.
- The appendices hold the glossary, the exemplar in numbers and the index of the canvas boards.

Czech words in italics are terms a person sees on AIA's screens or in the report. English words in `code` are files, modules or plan chunks in the repository. "The review" is the product owner's methodology review of 5 November 2026 (ADR 0046 § The methodology review, R1–R10, K1–K6). "The answer key" is `prototypes/case-chemvitro/ANSWER-KEY.md`. "The scope" is `MVP-SCOPE.md`, session A's document, whose wording this document takes as the source for the perspectives, the chunk ids and the acceptance A1–A6; where the two disagree, the scope wins and the coordinator is told. "The owner" is the product owner; "the methodologist" is the methodology owner, A. Bahbouh.

> **Three things this document does not do.** It does not invent a number: every ChemVitro figure comes from the exemplar's tables or the case pack. It does not decide what waits on the owner or the methodologist; chapter 13 lists those items. It does not describe how a screen works: the boards do that.

CHAPTER 1

The spine

A type-2 study is eight steps. The client's rule is written at the first and applied, as written, at the sixth. Everything between them exists to make that application honest. The canvas's navigator shows nine screens: the outline and the report are one step, because the report's structure is built once and printed once (the scope, § 1).

Step	Goes in	Comes out	The check	In the exemplar
1 ***Zadání***	the decision, its options and constraints, who decides and when, the client's files	the rule with its conditions, frozen and dated; interpretations recorded	frozen before any comparative result; every condition has a threshold and a measure	eight sites and the buy baseline; the CFO's rule of 2. 11. 2026 with four conditions; one interpretation waits for the CFO (R2-P02)
2 ***Data***	every client file and document	a contract per file; a findings table with a disposition per finding	units, currencies, price years, duplicates, missing, outliers, quotes that do not cover the same thing	42 findings; the key one is the Hanshin price (R2-P04 § 3.4, P1)
3 ***Rešerše***	what the files cannot give: public prices, rates, taxes, limits, law	typed public inputs, each with a quote, a source, a date and a state	verified at the source, or marked *prozatímní*	16 of 54 inputs public and unverified; *Stav podkladů* says so on page 1 (R2-P01)
4 ***Předpoklady***	the contracts, the public inputs, the client's answers	one table of inputs: low, likely, high, distribution, source, grade, guarantor, state	estimated from client data where possible; the scope of every estimate checked; correlations declared with a stability test fixed in advance	54 inputs (T9); overrun and delay from 36 comparable plants, drawn together (R2-P05)
5 ***Model***	the input table, the options, the timeline	the base case and the seeded futures; the same bytes on a rerun	one model for every option; a declared stopping rule; one change at a time in the extra computations	cost of supply 2027–2044; 20 000 futures, seed 20270201 (R2-P04 § 3.3, Appendix B)
6 ***Výsledky***	the futures and the base case	the rule applied as written; what could change the verdict; value of information; finance, operations, schedule	the verdict shown as a boundary; no probability outside the model; information valued for the rule first	no site passes; the flip at 14/15/16 % compensation chance; EVPPI second (R2-P06–P11)
7 ***Sociomapa***	the options' results across the futures, or their weighted scores across scenarios	a map with a mode contract, its relation matrix and what it may claim	links only where the contract allows; a stability test declared in advance; relative apart from absolute	the futures map failed its stability test and the report says so (R2-P12, T7, T8)
8a ***Osnova zprávy***	every claim the report will make	an outline ordered by the argument; each point with its source or method and why it matters	no point without a source; a point that is a judgement is marked as one	the part ids R2-P01…P18 and the grade legend (R2-P04 § 3.5)
8b ***Zpráva***	the closed outline	the report: summary, chapters, limits, literature, appendices, tables	every number with a grade in small type and a numbered source at the chapter's end; a validity review before approval	44 pages, seven grades, *Meze studie* admits what is unverified (R2-P15)

1.1 The shape of each step chapter

Chapters 2 to 9 use seven parts, always in this order and under these names, so that two steps can be compared line by line.

Part	What it answers
What goes in	The inputs the step takes, and from which earlier step or from the client.
What comes out	The artefact the step produces, as the next step and the report read it.
Checks	What must hold before the step is closed. A check is a refusal or a record, never advice.
How the exemplar did it	The ChemVitro report's own practice, cited by part id and table.
In AIA today	The code on `develop` the step reuses, by module. Nothing listed here is type-2 code; it is what the pivot stands on.
New	What has to be built, with the board and the plan chunk that build it, as the scope names them (`plan#CHUNK`; a chunk marked **new** exists in no plan file yet and the scope defines it in one line).
ChemVitro acceptance	What the answer key planted for this step, which of the scope's acceptance lines A1–A6 it serves, and which of the review's criteria K1–K6 it answers.

1.2 The scope's acceptance and waves

The scope (§ 6) makes the answer key the test, in six lines: A1 the three traps priced; A2 the EPC exclusions, the overrun from the comparables, spot purchases while late, tax reliefs out of the pre-tax rule, the kilogram and USD rows read right; A3 the rule applied as written at 8, 6 and 10 %, with the boundary printed where a declared judgement moves the verdict; A4 the renewal and Hanshin prices named as the assumptions most worth checking; A5 the report's structure R2-P01…P18 with every number graded and sourced and the tables from the same computation; A6 the same bytes on a rerun and a changed input refused without re-running downstream. Its waves (§ 7) build the steps in order over ten weeks: W1 *Zadání a Data*, W2 *Předpoklady a Model*, W3 *Budoucnosti a Rozhodnutí*, W4 *Informace a mapy*, W5 *Zpráva a přijetí*, each ending in a slice that runs on the ChemVitro files with nothing paid.

1.3 The rules the whole spine keeps

- **The client's rule decides.** It is written before any result and judged as written (K3). Anything else the study computes is evidence for "what could change the conclusion", not for the recommendation (review R3, R10).
- **A number is a reference to its origin.** It carries a source, a method, an uncertainty and a validation status (review I3). A number the origin does not support is not printed.
- **Unknown is never good.** A missing input keeps a range; a gate that is unknown blocks the recommendation until it is evidenced (review R2). A probability is a share of the model's futures, never a claim about the world.
- **Models propose, code computes, a person approves.** AIA's standing rule (ADR 0046 decision 1). A language model drafts text and suggests ranges; every number in the report comes from a program that gives the same bytes on a rerun.
- **A method the register does not hold stays provisional.** Its results print *odhad (metoda čeká na schválení metodikem)*, and a recommendation that rests on it stays *předběžné doporučení* (review R9, proposed; chapter 12).
- **Changes propagate.** A revised input marks every downstream result and approval stale until someone reviews it (K4). Every extra computation changes one assumption at a time and says so (R2-P15).

CHAPTER 2 · STEP 1

Zadání: the decision and its rule

The study begins by writing down what will be decided, between which options, under which constraints, by whom and when, and by which rule. The rule is frozen before anyone sees a comparison of the options.

2.1 What goes in

- The client's brief, as typed or pasted: one field of free text and the files that come with it (the case pack's brief is in `CASE.md` § 3).
- The decision-maker and the date the decision is taken, and the date the study must be ready.
- The options, each with the client's source for it, and a baseline every option is compared with. For ChemVitro: eight sites from the agencies' offers and "keep buying" from the supplier's letter.
- The constraints every option must meet (investment ceiling, start date, one plant at one site).
- The client's answers to the study's questions. The case pack fixes them in `CASE.md` § 8; where an answer is "we don't know", the input becomes a judgement with a range and says so.

2.2 What comes out

- **The decision record:** what is decided, why now, the goal, who decides, by when. The exemplar's § 1.1–1.2 (R2-P02).
- **The options and the baseline**, named as the client names them, with the source of each.
- **The rule**, frozen and dated, in the client's words: the measure it reads, the choice it makes, and each condition with its threshold. ChemVitro's: the highest median saving against buying, NPV of the cost of supply 2027–2044 at 8 %, if P(beats buying) ≥ 0,75, P(loss > 20 mil. EUR) ≤ 0,10, P(production by 1. 7. 2030) ≥ 0,80 and investment ≤ 175 mil. EUR; else recommend nothing and name the assumption most worth checking (R2-P02 § 1.3).
- **Interpretation records.** Where a condition does not say how it is measured, the study writes its reading down, the alternative reading, what each would change, and who confirms it. The exemplar's condition (4) names no probability; it is judged on the base case, the stricter reading "in every future" would fail all eight sites without changing the verdict, and the CFO is asked to confirm (R2-P02 § 1.3).
- **The views the client asked for beside the rule:** after-tax NPV, the verdict at 6 % and 10 %. They are shown; the rule does not read them.
- **The analyses the report will carry** (board Z4) and the perspectives that apply to the case (chapter 11).

2.3 Checks

- The rule exists before the first comparative result, and the freeze is dated and attributed. Reading the client's files at intake is feasibility work and is recorded as such (review I2).
- Every condition has a measure the model can compute and a threshold. A condition without one gets an interpretation record, never a silent default.
- An amendment of the rule after the freeze is approved by the decision-maker, dated, and printed in the report (board Z3, *zámek*).
- Every option is described comparably: the same plant, the same horizon, the same discount rate (review R1, question 1).
- The brief is Class B material by rule (`domain/material_rules.py`); a language model may read it, a client's dataset is Class A and is not sent to a model.

2.4 How the exemplar did it

The report states the decision, the eight sites, the fixed plant (40 000 t a year, 95 FTE, one process design) and the dates: the board meets on 29. 1. 2027, the site is decided on 1. 2. 2027, production by 1. 7. 2030, investment at most 175 mil. EUR (R2-P02 § 1.2). It prints the rule in full and says why its date matters: "that the rule was written before the computation prevents the result from being justified afterwards by a rule that would suit it" (§ 1.3). It records the one interpretation, prices the alternative reading and asks the CFO. It also notes that the rule reads the pre-tax NPV and that after-tax figures and other rates are shown at the client's request.

2.5 In AIA today

- The study's working copy with revision-checked saves and the operation log (`application/working_copy.py`, `domain/edit_ops.py`, `infrastructure/study_workspace_repository.py`), and immutable Design Revisions (`domain/design.py`, `infrastructure/study_design_repository.py`). The frozen rule is a revision of this kind.

- The Brief stage of the research workbench (`apps/web/src/components/rehome/research/`, `BriefStep`) and its materials (`domain/brief_materials.py`), reused as the screen the brief is typed into.
- The Lead, which triages a brief and proposes edits a person accepts (`application/research_lead.py`, ADR 0026, ADR 0041).
- The approval ledger with its closed set of subjects (`domain/approval_ledger.py`): the rule's freeze and every amendment become ledger rows.
- Material classes by rule (`domain/material_rules.py`, ADR 0008).

2.6 New

- A study kind for a business decision, `StudyKind.BUSINESS_MODEL` (ADR 0044; `cost-model-study#CP3`), and the business *Zadání* as data on boards Z1–Z7: the eight parts (`decision-brief#DB0`), triage as a Lead turn (`#DB2`), the decision core and the rule (`#DB3`), readiness R1–R7 (`#DB4`), the hand-off naming the files each step takes (`#DB5`).
- The rule as a typed object of n conditions, each with its own threshold and, where stated, a probability, judged per option at the stated rate and two alternates, frozen as a Design Revision (`decision-brief#DB3b`, **new**, extending `cost-model-study#CR7`, which judges one quantity against one threshold). Board Z3 § 03b (session B) shows it for ChemVitro: each condition with its threshold and what it reads, the four conditions named as gates that no saving outweighs, and the lock line (frozen 2. 11. 2026, amendments 0, interpretations 1, awaiting confirmation).
- The interpretation record a named person confirms: what the rule does not say, the chosen reading, the other reading and its effect, who wrote it and who confirms, where the report prints it (`decision-brief#DB3b`; board Z3 § 03b, *Záznam výkladu*).
- The model's parts as data on board Z5 § 04b (session B): the target quantity, the baseline, the futures, the shared timeline, what is in and outside the model, the investment ceiling as a gate, the result views the model will feed, and the perspectives that have components in the case.
- The perspectives chosen for the case (chapter 11), each with its components, in *Zadání* rather than in the model.

2.7 ChemVitro acceptance

- The rule is the CFO's of `CASE.md` § 6, with all four conditions and the "recommend nothing" clause, dated before any result (MV0 M-4; K3).
- The investment condition has an interpretation record, and the record names the CFO as the person who confirms it.
- The baseline is "keep buying" at the supplier's terms, not a site.
- The study asks the client only what `CASE.md` § 8 answers; anything else is a judgement with a range, labelled *úsudek*.

CHAPTER 3 · STEP 2

Data: the client's files under contract

Every client file is read, checked and bound by a contract before a number from it reaches the model. The checks produce findings, and every finding gets a disposition the report can print.

3.1 What goes in

- The client's data files: for ChemVitro ten files in `data/`, from monthly purchases to air-quality measurements (`CASE.md` § 9). The worker survey belongs to type 3 and is not read.
- The client's documents: seven PDFs, from the board's brief to the consultant's memo on air permitting. Each number taken from one is cited with its page.
- The decision record from step 1, which says what each file is for.

3.2 What comes out

- **A data contract per file:** which columns are used and how; units, currency and price year per column; which columns are personal and never leave the analysis copy; the period covered (board MV1; `client-data-in-analysis` chunk 2).
- **A quality report per file**, accepted by a person before use: missing, duplicate and out-of-range counts per used column over every row (`client-data-in-analysis` 2a).
- **The findings table with dispositions.** Every finding of the checks, with one of four dispositions: used; corrected, with the reason; left out; taken over as an assumption with a range. The findings that change the result or its reading go into the report (the exemplar's P1, R2-P18 Appendix B).
- **A document register:** each document by the name the report cites it under and its file name (the exemplar's Appendix A).

3.3 Checks

The checks are the exemplar's list (R2-P04 § 3.4), made explicit:

Check	What it looks for	ChemVitro instance
Units	a row whose unit differs from its column's	July 2025 volume 2 844 000: kilograms, read as 2 844 t (P1)
Currencies	mixed currencies in one column; a currency column that contradicts the header	comparable plants in EUR, CZK and PLN; one purchase row in USD (answer key § 3)
Price years	amounts in different years' prices without a year	one comparable plant without a price year (P1)
Duplicates	the same entity twice	one comparable plant entered twice; counted once, 36 plants (P1)
Missing values	cells the contract expects and does not find	no start date of the energy price fixation; waste prices only to 2028 (R2-P01 *Stav podkladů*)
Outliers	a value outside the range the file itself declares	one plant with capacity 91 kt against 18–60 kt; used only for overrun and delay (R2-P15)
Scope of a quote	quotes that do not cover the same thing	a cheaper PL2 waste quote covers only non-hazardous waste; PL1's rail rate assumes a spur that does not exist (answer key § 3)
File against brief	a number the brief states and the file contradicts	"about 1 470 EUR/t" against a 2026 average of 1 694 and 1 868 in September 2026 (P1)
Personal data	a column holding a person's identifier	none in type 2's files; the survey's respondents are pseudonymous and belong to type 3

A contract that names a column the file does not hold, or a quality report the person has not accepted, blocks the file from the model (MV0 M-1).

3.4 How the exemplar did it

The model uses every client file except the worker survey and all seven documents (R2-P04 § 3.4). The checks produced 42 findings, each with a recorded decision; the ones that change the result are in table P1 (Appendix B): the Hanshin price modelled as a range 1 470–1 694–1 868 EUR/t; the kilogram row; the single spot purchase of February 2026 turned into a spot premium of 20–60 %, most likely 39 %; the duplicate, the missing price year and the 91 kt outlier among the comparable plants; consumption growth of 4 % in the data against the client's 1–3 %, the client's range used; raw-material freight not added because the raw-

material price is delivered to site; Odrzyca's rail quote and Eschfeld's bridge. Twenty-two inputs the files do not hold became judgements with a range. A separate document, *Kontrola dat*, holds the contracts and all 42 findings (Appendix C); the report prints the ones that matter.

3.5 In AIA today

- A client's sources kept once per client and SHA-256, read by a format reader into segments with locators (`domain/client_sources.py`, `infrastructure/source_reading.py`, `infrastructure/client_source_repository.py`): PDF by page and table row, XLSX and CSV by sheet and row, DOCX by paragraph.
- The dataset preview: per column the header, kind, fill, distinct values and samples, with personal columns detected and masked (`domain/client_datasets.py`; patterns pinned in `fixtures/client_datasets/personal.json`).
- The intake sheet, *Přidat podklady*: each file kept and read at once, one knowledge item per file (`apps/web/src/components/aia/clients/IntakeSheet`).
- Document text with ZIP bounds and a bounded table reader (`infrastructure/document_text.py`).
- The evidence policy for client data: client-reported evidence under an accepted contract (ADR 0027, ADR 0037; `client-data-in-analysis` 8a built).

3.6 New

- The data contract as a type and the route a person accepts it through (`client-data-in-analysis#2`), the quality report (`#2a`), the analysis copy of named columns and its accessor (`#3`, `#3a`, `#3b`), admission as *z dat klienta* (`#5`), the contract accepted on the Data step (`#10`). Boards MV1 § 01b–03b and M1 § 01b–03b (session B) show it for the purchases file: the contract per column with the check note, the quality report with a disposition per row, what the model gets from it (summaries, never rows), and the 42 findings with the seven that change the result.
- A number from a document proposed with its page and checked literally on that page (`client-data-in-analysis#11`); an estimate from client rows as a Metodika method (`#12`).
- Declared conversions (unit, currency at a dated rate, price year), each a finding with its disposition, and the P1 table of findings with the four dispositions *použito / opraveno / vynecháno / převzato jako předpoklad* (`client-data-in-analysis#13`, **new**; board M1 § 03b). `cost-model-study#CR1` covers reconciliation, period coverage, currency and robust outliers, accepted on record.
- The document register with cited names (the exemplar's Appendix A; part of `decision-brief#DB5`'s hand-off).

3.7 ChemVitro acceptance

The answer key planted these in the files (§ 3). A study that misses one has a gap in its checks, whatever its verdict.

- `srovnatelne_zavody.xlsx`: P-17 twice, the second with a rounded investment; P-23's capacity 91 in t/day, not kt a year; P-31 without a price year; investments in EUR, CZK and PLN in each plant's own price year.
- `nakup_vtx40_2023-2026.xlsx`: the 2025-07 volume in kg; the 2024-03 price in USD; an extra 2026-02 row from another supplier, the spot price for a late start. The key names 2 180 EUR/t and the exemplar 2 320 against 1 672 (P1); both are the same row read two ways, price alone or price plus freight (2 180 + 140 against 1 576 + 96), as board MV1 § 02b shows. The acceptance reads the file and says which it prints.
- `trh_prace_regiony.xlsx`: wages gross, in local currency, without employer contributions.
- `odpady_nabidky.csv`: PL2's second quote covers only non-hazardous waste.
- `doprava_nabidky.csv`: PL1's rail rate assumes a spur to the site.
- `hodnoceni_lokalit.csv`: one rater scored with the scale reversed and says so in a note; the consultant adds about 2,5 points to the two sites its agency brokered; two cells are missing. Type 2 does not use this file in its model; if it draws an expert map, the map must show these.

Criteria: K6, claim integrity, where a table and the text disagree about a reading; K4, since a corrected finding propagates to every result that read it.

CHAPTER 4 · STEP 3

Rešerše: public inputs with a state

What the client's files cannot give, public sources give: prices, rates, taxes, limits and law. Each is a typed input with a quote, a source and a date, and it stays provisional until someone has verified it at the source.

4.1 What goes in

- The list of public inputs the model needs, from the decision record and the contracts. For ChemVitro, `CASE.md` § 10: the EU ETS allowance price and its outlook and whether a 24 MW boiler falls under the scheme; air-quality limit values and the permitting rules of three countries; hazardous-waste fees; wages, employer contributions and unemployment; corporate income tax, depreciation and the nature of each incentive; industrial energy price outlooks; exchange rates.
- The client's data posture (*Pravidla*, `domain/client_rules.py`) and the study's budget, which bound what a priced route may spend.
- Findings that frame the problem: the literature the report cites (R2-P03), found and quoted the same way.

4.2 What comes out

- **Typed public inputs** (board MV2): a value with its unit, the quote it was read from, the source and its date, the kind of source (*Úřední / Oborový / Ostatní*), the grade (*veřejný zdroj* or *právní předpis*) and the state: *ověřeno* once checked at the source, else *prozatímní*.
- **Findings by analysis** (board MV2b): PESTEL, Porter or benchmark findings as proposals a person accepts, each a quote in a captured source.
- **The literature review** as the report prints it: for each work, what it says and what follows for this report (R2-P03, five works).
- **The state line for the report's first page:** how many inputs are public, how many unverified, and whether the unverified ones could change the verdict (R2-P01 *Stav podkladů*).

4.3 Checks

- A finding is a quote in a captured source, or it is nothing (ADR 0017). A value without a quote and a date is not an input.
- A query's data class is inherited from what it was written from and never lowered by its words (ADR 0017, `domain/deep_research/classification.py`). A client's dataset never leaves as a query.
- A provision is cited; no legal advice is given (`CASE.md` § 10, R2-P04 § 3.3).
- An unverified input is marked *prozatímní* in the input table, in the chapter that uses it and on page 1. Verification at the source is a dated act by a named person and changes the state, nothing else.
- A public input the exemplar's method tests for sensitivity says whether its range changes the verdict (chapter 7).

4.4 How the exemplar did it

Sixteen inputs are public (the text says "16 of 54"; T9 has 67 rows, a claim-integrity finding recorded in chapter 5): the allowance price (89,10 EUR/t on 28. 10. 2026, Investing.com, unverified), its growth from December 2030 futures, free allocation, energy price volatility and long-term movement, three price correlations, the CZK and PLN rates, employer contributions in three countries, inflation (T9). The report says, on page 1, that the authors could not verify them at the source because access to public sources was closed while the report was written, that the amounts may change after verification, and that the correlations and the energy trend do not change the verdict while stricter 2030 air-quality limits could (R2-P01). The tax rates, depreciation, the form of incentives and Directive (EU) 2024/2881 are likewise provisional; the incentives are read correctly as tax reliefs in Czechia and Poland and a cash grant in Germany (R2-P10). The literature chapter cites AACE 18R-97, Flyvbjerg, Merrow, Vose, Strong and Oakley, Williamson and Lempert, Popper and Bankes, each with the consequence for the report (R2-P03).

4.5 In AIA today

- Deep Research, frozen since #166: plan, investigate, merge, verify, synthesize, publish; grounding of every quote in a content-addressed snapshot of the same track (`domain/deep_research/`, `grounding.py`, `web.py`, `sources.py`); the sealed bundle.
- The one way a query or URL leaves: `application/web_retrieval.py` `RetrievalGate` (classify, egress, metering, reserve, journal, call, journal). Live routes: Czech and English Wikipedia, priced Brave, Common Crawl (`aia_executors.deep_research_runtime`).
- Design Research and Interpretation Research as the two purposes; a finding never writes a design or a deterministic artefact (ADR 0021).

- A captured source a person checked becomes a SOURCE knowledge item, *Ověřeno, přidat zdroj* (`application/verified_sources.py`).
- The Lead's web lookup, accepted and not built (ADR 0040).

4.6 New

- Typed inputs (value, unit, period, range, source, as-of date) extracted by code from accepted quotes and from structured routes, with the state *prozatímní / ověřeno u pramene* (`deep-research-v2#DRV1` the typed input, `#DRV2` the extractor, `#DRV3` the extraction step); credibility tiers *Úřední / Oborový / Ostatní* (`#DRV5`); as a provenance of a model input, `cost-model-study#CP1`'s `public_source`. Board MV2 § 01b (session B) lists the sixteen provisional rows of T9 with value, unit, source, date, grade and state, and § 02b makes verification a step with three outcomes: Deep Research finds and cites (*prozatímní*), the researcher opens the source and records the date and their name (*ověřeno u pramene*), what no source holds stays a judgement with a guarantor.
- Structured public routes: SDMX for Eurostat and the ECB (`public-data-routes#PDR1`), ČNB rates (`#PDR2`).
- A legal point as a verbatim provision that becomes a parameter (tax rate, depreciation, contributions): e-Sbírka and EUR-Lex routes, the legal point, legal parameters (`legal-context#LGL1`, `#LGL2`, `#LGL3`).
- The first-page state line composed from the input table's states (part of the report-2 recipe, `research-findings-stage#FND13`, chapter 9).
- Findings by analysis (PESTEL, Porter, benchmark) as proposals, each a quote (board MV2b, unchanged in this round).

4.7 ChemVitro acceptance

- Every public value in the model carries a quote, a source, a date and a state; none is invented (`CASE.md` § 11).
- The 24 MW boiler is above the 20 MW threshold: carbon is a cost at every site (answer key § 3, memo p. 1).
- Incentives are read by their nature: the Czech investment incentive and the Polish CIT exemption are tax reliefs and do not change the pre-tax figure the rule reads; only the German GRW is a cash grant. Subtracting PL2's "CIT exemption up to 50 %" from the investment is a misreading (answer key § 3).
- The revised directive (EU) 2024/2881 with its 2030 limits is noted as a caveat, as the exemplar does; it is not planted and earns credit, not a requirement.
- Criteria: K2, where a public range is a decisive input; K5, where verification at the source has a date that the plan of action depends on.

CHAPTER 5 · STEP 4

Předpoklady: one table of inputs

Every input of the model is one row of one table: its low, likely and high value, its distribution, its source, its grade, the person who stands behind it and its state. The table is the contract between the data and the model, and the report prints it whole.

5.1 What goes in

- The data contracts and the findings table from step 2; the typed public inputs from step 3; the client's answers from step 1.
- The comparable cases the client holds, from which ranges can be estimated rather than judged: for ChemVitro, 36 comparable plants with their overrun, delay and what each built beyond the plant.
- The scope statement of every estimate the client supplies: what the EPC contractor's figure includes and excludes (`epc_odhad.pdf` p. 3).

5.2 What comes out

The input table, the exemplar's T9 (R2-P18). Its columns are the method:

Column	Rule
Vstup, jednotka	one input, one unit; a fixed parameter is a row like any other
Dolní · nejspíš · horní	three values; equal for a fixed value. What the ends mean is written down for every decisive input (review R6): a 90 % interval of an estimate, contractual bounds, the worst and best credible case
Rozdělení	PERT by default for a range a person stands behind; the estimate's own distribution where it comes from data; an empirical draw from comparable cases; yes/no with a probability for an event; a mixture where experts disagree; a fixed value (*pevná hodnota*)
Zdroj	the file and row, the document and page, the public source and date, or "not in the files"
Stupeň	one of the grades of chapter 12: *z dat klienta*, *odhad*, *úsudek*, *veřejný zdroj*, *právní předpis*
Stojí za ním	the guarantor: a named role in the client (the CFO, head of purchasing, head of EHS) or the author. Every assumption has an owner (the review's "to keep")
Stav	*ověřeno* (from client data), *prozatímní* (public, unverified) or *úsudek* (the client has no answer)

Beside the table: **the correlation record** (which inputs move together, with what coefficient, why, and the stability test their judgement will be put to), **the scope check** of every supplied estimate against each option's documents with the exclusions priced, and **the stress cases** outside the ranges, kept apart from the futures (board M2 § 04).

5.3 Checks

- Where the client's data allows an estimate, the range comes from the data, not from a judgement; the row says which. The exemplar: overrun and delay from the comparable plants, not from the estimator's ±25 % (R2-P05).
- Two things that happened together in the data are drawn together in the model. Overrun and delay have a Spearman correlation of 0,89 across the comparable plants, so each future draws one real plant and takes both (R2-P05).
- Every supplied estimate is checked for scope against each option's documents, and each exclusion is priced or stated as missing (R2-P05, fourth assumption; answer key § 3: a study that adds nothing to the EPC figure has missed all three traps).
- A decisive input has an uncertainty type: natural variation, parameter, expert disagreement, model structure, or a severe event outside the range; the last goes to a stress test, not into the distribution (review R6; board M2 § 02).
- Experts who disagree from the same documents are a mixture that does not narrow the range; the report says they differ in reading, not in facts (board M2).
- A correlation that is a judgement is declared as one, with the alternative settings it will be tested under, before anything is computed (R2-P05 fifth assumption, Appendix B).
- No input the files do not hold is given a point value. It gets a range and the grade *úsudek*; the exemplar has 22 such inputs (R2-P18 Appendix B).

5.4 How the exemplar did it

Chapter 4 (R2-P05) names the two assumptions the verdict rests on that no document supports: the 20 % chance that an authority imposes compensation measures where the station is under the limits, and the terms of purchase after 2035. Then three more: overrun and delay from 36 comparable plants (median overrun 17 %, 90th percentile 30,2 %, maximum 61 %; delay median 3

months, maximum 15), drawn jointly; the exclusions of the EPC estimate priced per site from the site's documents (18 ha of land everywhere; the bridge at Eschfeld 3,2–5,0 mil. EUR; the spur at Odrzyca 3,4 km at 1,9–2,8 mil. EUR per km; pre-treatment at Lipová and Lindenau 6,7–14,1 mil. EUR; compensation 3–15 mil. EUR where imposed; the park fee at Wendelstedt 1,1 mil. EUR a year); and the energy price correlations as judgement (0,8 electricity between countries, 0,6 electricity–gas, 0,4 gas–allowance), with the stability test declared in Appendix B. The base case already carries the median overrun. The report's text says the model has 54 inputs and that 16 of 54 are public; the table T9 has 67 rows (29 *ověřeno*, 16 *prozatímní*, 22 *úsudek*). Session B counted from T9 and recorded the difference on board M2 § 01b as a finding for the report; it is the kind of contradiction between text and table that K6 exists to catch, and this document counts from T9.

5.5 In AIA today

- Nothing holds an input table yet. The nearest ancestors are the simulation core's frozen WorldModel with reject-not-clip validation (`domain/simulation/`) and the research template's readiness rules (`domain/research_design.py`), which show the pattern: a typed object, validated, pinned to a run.
- Guarantors and audit: the approval ledger (`domain/approval_ledger.py`) and the audit trail every admin route writes.
- Methodology as data the methodologist owns: versions, activation and bounds per editable field (`domain/methodology/`, ADR 0023). The distribution defaults and the stability thresholds belong there.

5.6 New

- The input as a platform type with provenance (`cost-model-study#CP1`: `client_dataset`, `stated`, `public_source`) and low / base / high with a guarantor on every input (`#CR2`); the input table as the stage *Předpoklady*. Board M2 § 01b–03d (session B) shows it for ChemVitro: T9 with a filter by state, the decisive rows marked; the uncertainty type, the meaning of the ends and the reason for each decisive input; the estimate from the 36 plants; the scope check as a table of excluded items with where, by what document, price and months, grade; and the correlations as judgement with the stability test and the rule's re-application in three settings written down on 5. 11. 2026, before the computation.
- Distributions and the correlation record: PERT, Iman–Conover rank correlations, a non-PSD matrix refused (`#CR3`); empirical joint draws, one row of a client table per future carrying several inputs together, the table pinned by SHA-256 (`#CR3b`, **new**); yes/no events with a probability and fixed and uniform values as T9 names them. A mixture for expert disagreement (review R6; board M2 § 02 for Velis) is on the canvas and in no chunk yet; board MV5 lists it among what the methodologist signs for the Monte Carlo entry.
- The estimate from comparables as a Metodika method (`client-data-in-analysis#12`; the exemplar's third assumption).
- The scope check: what an estimate's document excludes against what each option's documents require, each exclusion an input with its range (`cost-model-study#CR1b`, **new**).
- The stability test declared before computing, as the futures map's chunk defines it (`sociomapa-systemu#SYS9b`, chapter 8).
- The state column and its three values, read from steps 2 and 3 (`deep-research-v2#DRV1`'s state for public inputs).

5.7 ChemVitro acceptance

The answer key's three traps live in this step. Each is a cost the brochure hides and a site document reveals; the study must find and price it.

Trap	Where it is	What the exemplar priced
Wastewater, CZ2 Lipová and DE2 Lindenau	`vyjadreni_cov.pdf` p. 2 and 5: the sewer bylaw's sulfate and COD limits against the process effluent (1 800 and 2 400 mg/l); Lindenau's plant has spare capacity for 250 m³ a day against 430 needed	pre-treatment 6,7–14,1 mil. EUR at both, from 8 comparable plants; Lindenau's direct-discharge permit 0–9 months (T9)
Air, CZ1 Vrbice and PL2 Bory	`ovzdusi_stanice.csv` 2025 values over the limits, combined with `memo_povolovani_ovzdusi.pdf` p. 2–3: compensation measures and 9–15 / 12–18 months longer permitting	compensation with probability 0,8, 3–15 mil. EUR, 9–18 months (T9); both sites then miss the deadline in most futures (T1)
Logistics, PL1 Odrzyca and DE3 Eschfeld	`vyjadreni_doprava.pdf` p. 7: the investor builds the 3,4 km spur and buys the land; p. 6: the bridge carries 16 t, tankers weigh 40 t	the spur at 1,9–2,8 mil. EUR per km plus road haulage until built; the bridge 3,2–5,0 mil. EUR and 12–30 months to heavy-traffic approval (T9)

- Overrun from the comparables, median 17,0 %, not the estimator's band; the duplicate removed first (answer key § 3).
- The decoys left alone: Kamenec's single 2021 exceedance (under the target since); Eschfeld's plant charges from COD 2 500 mg/l against 2 400 in the effluent; Lipová's turning lane at about 2 mil. Kč; Bory's municipal plant does accept the effluent.
- Wendelstedt's park fee of about 1,1 mil. EUR a year, named only in the wastewater statement (p. 4).
- Criteria: K2, uncertainty fitness, is this step's criterion: decisive distributions, dependence and stress cases justified. K1's second question (material costs and dependencies in the model) is answered by the scope check.

CHAPTER 6 · STEP 5

Model: one computation for every option

The model computes the measure the rule reads for every option in the same futures. It is code, pinned by version and seed, and a rerun gives the same bytes. Nothing in it is a forecast or a legal opinion.

6.1 What goes in

- The input table of step 4, frozen as a revision; the options and the baseline of step 1; the rule's measure.
- The timeline every option shares: for ChemVitro, decision and permit applications on 1. 2. 2027, permitting 8–11 months plus the site's extensions, construction 22 months plus the drawn comparable plant's delay, ramp-up at 60–90 % for six months (Appendix B).
- The seed and the number of futures, or a stopping rule that sets the number.

6.2 What comes out

- **The base case:** one computation with every input at its likely value. It compares the options and feeds the sensitivity analyses; it is not an estimate of what the client will save (R2-P10).
- **The futures:** N seeded draws of all inputs at once, with their correlations, each future computing the baseline and every option. From them: the median, the 90 % interval, and every probability as a share of futures (R2-P04 § 3.3).
- **The extra computations** on the same model and the same futures, each changing one assumption: the decision boundary over an assumption's range, delay of the decision, hiring delay, alternative terms, correlation settings (Appendix B).
- **The run record:** model type and version, input revision, seed, N, the methods pinned, the artefacts with their SHA-256 (MV0 M-2, M-3).

6.3 Checks

- Every option is computed in the same futures by the same model; the baseline is one of them (R2-P04 § 3.3).
- The same inputs and seed give the same bytes (M-3). This verifies the implementation, not the model's fitness for the decision (review R1; chapter 13).
- The number of futures follows a stopping rule tied to the decision: the verdict of each condition is known to within a stated margin (board M2: ±0,5 pp on the pass probabilities) or N is declared and justified. The exemplar declares N = 20 000 (R2-P04 § 3.3).
- Cash flows are discounted at the rule's rate, nominal or real as declared, with the convention named (mid-year, inflation 2 % provisional; Appendix B).
- Tax is inside the model as a fixed parameter per country, so the after-tax view agrees with every other view (review I4). The rule reads the pre-tax figure.
- What is outside the model is listed where the model is described, not discovered in the limits (R2-P15: transfer prices, hiring time in the schedule, raw-material freight, the annual step).
- An extra computation changes one assumption; combinations that are not computed are said not to be (R2-P15).

6.4 How the exemplar did it

The cost of supply is the investment, operations from start-up, VTX-40 bought until start-up, purchases above 40 000 t, less a residual value of 30 % of the investment in 2044; the saving of a site is the baseline's cost less the site's, in NPV, positive when the plant is cheaper (R2-P04 § 3.3). Purchases follow the supplier's formula to 30. 6. 2030, then the renewal premium, or the spot price while the plant is late (Appendix B). Operations are six items per tonne: raw materials at 540 EUR/t delivered, electricity and gas at the sites' offers moving with the country's price path, CO_2 at the allowance price less free allocation, water, sewer, hazardous waste and product freight at the quotes, 95 staff at the operator wage times a wage mix and contributions, maintenance and insurance at 2–4 % of the EPC estimate (Appendix B). The futures number 20 000 with seed 20270201. The base case carries the median overrun of 17 %; the futures draw one comparable plant for overrun and delay together. Eight extra computations reuse the same futures, each changing one assumption (Appendix B).

6.5 In AIA today

- Runs as workflows on the worker, with lease-fenced steps, artefacts, provenance and dependency edges, each artefact pinned by SHA-256 (`apps/worker`, `infrastructure/artifact_repository.py`, `domain/workflow.py`, `domain/pipeline.py` fingerprints).

- A run pins the organization's methodology in force and records it in its identity (`domain/methodology/pins.py`, `application/research.py`).
- The deterministic simulation core: a frozen WorldModel, reject-not-clip validation, seeded scenarios and frozen results (`domain/simulation/`). Its discipline is the model's; its content is not.
- Seeded synthetic data from `random.Random(seed)` and the Sociomap golden fixtures pinned by SHA-256: the rerun-equals-bytes practice (`packages/aia_core/tests/fixtures/sociomap/`).
- Budgets and cost ceilings per run, which a model computed by code does not need (a CALCULATED run has a known 0,00 USD ceiling, `cost-model-study` CP5).

6.6 New

- The platform: the model-type contract and the determinism harness (`cost-model-study#CP2`), the kind and the working copy (`#CP3`), the run with inputs frozen by artefact id (`#CP5`), the API (`#CP6`), the ESTIMATED grade with seed and method pinned and the Metodika module of defaults (`#CP10`). Board M3 (session C) shows the type for ChemVitro beside Velis: the shared timeline as one drawing (§ 02) and the eleven items of the cost of supply with their kind and base formula (§ 05), every number held to the exemplar by `tools/results_build.py`.
- **A model type of its own, `site_cost_of_supply`** (`cost-model-study#CM8`, **new**): a timeline per option whose dates are inputs, purchases, operations, residual value, tax as a fixed parameter, yearly cash flows, NPV before and after tax at a stated rate, T4–T6 as its results, and the convergence rule tied to the decision (the futures run until the rule's verdict is stable, ±0,5 pp). The planned type *scénáře nákladů* (`#CM1`–`#CM7`) has no timeline and no bought baseline, so the scope makes type 2 a second type rather than an extension.
- Monte Carlo with the input table's distributions and correlations (`#CR3`), the joint draw from comparables (`#CR3b`), PCG64 under the run's seed; board M3.
- The cash-flow schedule, NPV and discounted payback, and IRR by a stated root rule (`#CM7`; `quantitative-families#FIN1`, `#FIN2`).
- The extra computations as declared operations on a stored run, each one change at a time and said so: the boundary, delay and hiring tables, post-2035 and correlation recomputations (`cost-model-study#CR7b`, **new**, chapter 7; board M4).

6.7 ChemVitro acceptance

- Seed 20270201, 20 000 futures, the same seed gives the same result (`STUDIES.md` § 2, MV0 M-3).
- The spot price for a late start: any VTX-40 needed after 30. 6. 2030 and before the plant runs is bought at spot, as in February 2026 (`CASE.md` § 8).
- Tax reliefs out of the pre-tax figure the rule reads; after-tax at each country's rate (answer key § 3).
- Every option on the same timeline from 1. 2. 2027; a site's extra months (compensation, the bridge, the spur, the permit) enter its own schedule (R1 question 1).
- Criteria: K1's first and second questions (comparable options; material costs and dependencies); K4, a revised input invalidates the run's results and approvals.

CHAPTER 7 · STEP 6

Výsledky: the rule applied, and what could change it

The results answer four questions in order: what the rule chooses, how far the verdict is from flipping, what information would change it, and what the finance, operations and schedule of each option look like. The first question is the recommendation; the rest is evidence about it.

7.1 What goes in

The futures and the base case of step 5; the rule of step 1; the extra computations declared in step 5.

7.2 What comes out

Five views, each a set of tables the report reuses (the exemplar's T1–T6, R2-P06 to R2-P11):

View	Content	Exemplar
Rozhodnutí	the rule table per option (the measure's median and 90 % interval, each condition's value against its threshold, pass or fail), at the rule's rate and the client's other rates; the **decision boundary**: the rule's choice as one decisive assumption moves over its range; delay tables; stress tests; the tornado; break-even values; alternative terms; the correlation check	T1–T3; the flip at 14 / 15 / 16 %; delay 0/2/4/6 months with and without compensation; hiring delay; tornado; break-even prices; post-2035 terms; three correlation settings (R2-P06–P08); board M4 § 02–04
Hodnota informace	two measures, in this order: **the value for the rule** (what knowing an answer in advance and applying the rule again would bring, averaged over the possible answers), then **EVPPI** (the gain in mean saving from knowing one input); then **the enquiry table**: what would be learned, the output, the duration, the cost, and the consequence of waiting	122,5 mil. EUR for the rule at Odrzyca against 4,9 by EVPPI; four enquiries (R2-P09); board M4
Finance	pre- and post-tax NPV at the rates, IRR, payback, investment by item with the probability of staying under the ceiling; incentives by their nature	T4, T5 with P(investment ≤ 175) and its percentiles (R2-P10); board M4b § 02–03
Provoz	cost per unit by category in a full year of operation, base case	T6, 2032 (R2-P11); board M4d § 04, each column named as its component; freight, the spur, the bridge and the trucking surcharge on board M4f § 02–05
Harmonogram	the start-date distribution per option against the deadline (median, 10th–90th percentile), P(on time), and the effect of each month of delay	the start chart and the delay tables (R2-P07); board M4g (new, session C): the start against the deadline per site, what lengthens each tail, and decision delay × hiring delay as one grid whose empty cells the report did not compute

7.3 Checks

- The rule is applied exactly as written, condition by condition; the pass column is computed, never typed. The report prints the value of every condition against its threshold, pass and fail alike (T1).
- The verdict is given as a boundary, not a point: for the one or two assumptions the verdict rests on, the table of what the rule chooses at each value (R2-P06 § 5.3). A verdict with no boundary table has not been tested.
- A probability is a share of the model's futures under the inputs that produced it; every probability says which assumption it stands on (R2-P04 § 3.3). The report claims no probability for anything the model does not compute (R2-P15).
- The value of information is computed for the rule first and for the mean second; neither is a budget for the enquiry; the enquiry table carries duration and the consequence of waiting (review R7; K5).
- A median is not a mean: the rule's choice of statistic is kept, and the report says what the right tail does to the other (R2-P06 § 5.1).
- The base case and the futures may disagree; the report explains why when they do (R2-P08 § 7.1: break-even in the base case against loss in the median).
- Every extra computation names the one assumption it changes (R2-P15).

7.4 How the exemplar did it

No site passes at 8, 6 or 10 % (T1–T3). The Czech and Polish sites fail only the deadline condition (0,75–0,76 against 0,80 for the three leaders, 0,19–0,21 for Vrbice and Bory); the German sites fail mainly the loss condition (0,14–0,21 against 0,10). The verdict is a boundary: at a compensation chance up to 14 % the rule picks Odrzyca, at 15 % Kamenec, from 16 % nothing; the model uses 20 % (R2-P06 § 5.3). Delay of the decision by 2, 4 or 6 months lowers every P(on time); with compensation ruled out, the rule still picks Odrzyca after two months and nothing after four (R2-P07 § 6.1). Hiring delay is a stress test that adds to decision delay (§ 6.2). The tornado puts the 2026 Hanshin price first (a swing of 134,6–152,7 mil. EUR), the renewal premium second (57,7) and overrun third (34,2–44,5) (R2-P08). Break-even prices lie 1 249–1 308 EUR/t for Czech and Polish sites, 1 413–1 450 for German ones; at the 2025 average of 1 296 EUR/t the median of every site turns negative (§ 7.1). Post-2035 terms are the second lever: at a 30 % premium Wendelstedt would pass (§ 7.2). Three correlation settings move medians by at most 1,4 mil. EUR and probabilities by at most 0,01 (§ 7.3). The value for the rule of the authorities' answer is 115,2–152,1 mil. EUR; EVPPI of the same information is 4,9 mil. EUR, of all inputs 14,2 (R2-P09). Finance: base-case saving 170,7 / 165,9 / 159,9 mil. EUR for the three leaders at 8 %, IRR 18,2–20,5 % in Czechia and Poland, payback 2034–2036; after tax Kamenec and Odrzyca are level (T4). Operations: 1 290–1 311 EUR/t in Czechia and Poland against 1 434–1 463 in Germany in 2032, the difference in staff, maintenance and freight (T6).

7.5 In AIA today

- A number reaches a result only as an admitted claim (`domain/evidence/admission.py`, `application/admitted_results.py`); suppression is decided on the server and a suppressed row prints its reasons, never a number. The pattern holds for model results: a cell the method does not support is not printed.
- The results screens of a research run and their reading in words (`ExecutionSteps.tsx`, `lib/result-reading.ts`): unknown is never ordered, an interval missing means no order.
- The report's print grades and the evidence ledger (`domain/report/evidence.py`), the per-question tables as exports (`domain/report/table_export.py`, RD12).
- The budget line and spend reading as the pattern for "unknown never zero" (`lib/budget-line.ts`, `lib/run-spend.ts`).

7.6 New

- The rule judged as written (`cost-model-study#CR7`) and, new, the rule at three rates, the decision boundary over one input's range, delay and hiring tables, post-2035 and correlation recomputations, each one change at a time and said so (`#CR7b`, **new**); the method on screen (`#CR8`); grades ESTIMATED for everything from the futures and CALCULATED for the base case (`#CP8`, `#CP10`). Boards M4, M4b, M4d, M4f and M4g (session C) carry the ChemVitro tables beside the Velis sections, each result section naming the component it feeds in the scope's words with *v modelu / mimo model*.
- Value of information for the rule: the rule re-applied per answer branch, the average of what it then picks against today's "nic", and the enquiry table with its effect on the deadline (`#CR6b`, **new**; K5); EVPPI beside it (`#CR6`).
- Break-even and one-way sensitivity (`#CM2`), global sensitivity beside the tornado (`#CR4`), three chart kinds the exemplar needs and the model lacks: the absolute stacked bar, the tornado, the line with marked points (`#CP8`).
- Finance: scenarios on NPV and IRR (`quantitative-families#FIN3`), tax relief against grant as legal parameters (`legal-context#LGL3`); board M4b.
- Operations: cost per unit by category in a stated year (board M4d); *Harmonogram*, the start-date distribution per option against the deadline, P(*včas*) and the months-of-delay table (`cost-model-study#CM9`, **new**; board M4g, `KM12Schedule.dc.html`, new).
- Boards M4c *Účetnictví* and M4e *Trh* are parked for type 2: they have no component in the ChemVitro case. The Výsledky row now runs M4d → M4g → M4f → M4e.

7.7 ChemVitro acceptance

The answer key's reference model finds CZ3 Kamenec passing at 8 % (median +39,8 mil. EUR, P(beats) 0,82, P(loss > 20) 0,09, P(start) 1,00), CZ2 close behind, PL1 on the edge of the loss limit, nothing at 10 %, and names the renewal premium (12–22 %) as the assumption most worth checking (answer key § 1). The exemplar finds no site passing at any rate and names the compensation chance first and the Hanshin price second. The two differ on one judgement: the reference prices offsets only where a station exceeds the limits; the exemplar adds a 0,2 chance everywhere else in Czechia and Poland, and says that this judgement decides the verdict and has no document behind it (R2-P15).

- The acceptance therefore does not require the exemplar's verdict or the reference's numbers. The key says so itself: "judge a report by whether it found the planted facts and applied the rule honestly, not by whether it hit these numbers."
- It requires: the rule applied as written with every condition's value printed; the verdict given with its boundary on the assumption it rests on; the assumption most worth checking named with its value for the rule; the three traps and the overrun from the comparables visible in the investment and the schedule (CZ1 and PL2 fail the start date; DE3's P(start) falls with the bridge).

- Both the exemplar and the reference agree that the brochure favourite, PL2 Bory, cannot start on time. A study that recommends Bory has read the brochure, not the files.
- Criteria: K3 (the rule's choice kept apart from supplementary scoring), K5 (every enquiry states its effect on the deadline), K6 (the rule table, the summary and the boundary table say the same numbers).

CHAPTER 8 · STEP 7

Sociomapa: the futures map

A Sociomap places the options as points in a landscape: two options lie close when the same observers rate them alike, and the height shows the mean rating. In a type-2 study the observers are the model's futures. The map is evidence about the structure of the options' risks, never about the recommendation.

8.1 What goes in

- The futures of step 5 as a table: one row per future, one column per option, the cell the measure the rule reads (the saving against the baseline in that future).
- The mode contract for futures (board MV3 § 02, mode *Scénáře*): what a row is, what the claim is about, what is measured, what the null means, the resampling, what the stability measures.
- The stability test, declared before the computation: which alternative settings the map is recomputed under and what agreement counts as stable.
- The method pinned: `aia-sociomap-5`, the organization's methodology in force.

8.2 What comes out

- **The futures map** in two views: the 3D terrain and the top view that does not distort distances by perspective (R2-P12 § 11.2). Heights are mean saving; links are drawn only between pairs the method marks reliable; a warm link means co-movement, its width the relation's strength.
- **The relation matrix**, the exemplar's T7: each pair's relation across the futures, from which the map is drawn.
- **The shared-failure table**, T8: when an option is worse than the baseline, the share of those futures in which each other option is worse too. This reads the absolute, uncentred futures; the map reads the centred, relative ones.
- **The stability result:** the maps under each setting, the pairwise agreement of their distance orders, and pass or fail against the declared threshold.
- **The register status:** the map's method at its spec fingerprint, approved or *interní* (`domain/sociomap_register.py`), printed on the map and in the report.

8.3 Checks

- **Mode contract.** A row is one independent draw of the model. The claim is about the model and its assumptions, not about the market. Centring each future on the mean of the options makes the map show relative performance: who does better in the same future, not whether all do well (R2-P04 § 3.3; review R5).
- **Relative apart from absolute.** The question "is option B a backup for A" is answered by the uncentred shared-failure table, not by the map: centring removes the common level, which is exactly what shared failure is made of (R2-P12 § 11.3; board MV3 § 05).
- **Links only where the contract allows.** The exemplar drew links where the `-5` method's pair test, with the false-discovery correction, marked a pair reliable (2 of 28). Board MV3's scenario contract says a null makes no sense when the model creates the relations, and asks instead for stability across seeds and assumption changes, with no p-values. Which of the two applies to futures is an open question for the methodologist (chapter 13); until it is answered, the report prints which rule it drew by.
- **Determinedness.** Split-half agreement of two halves of the futures against the method's threshold says the layout is not an artefact of the draw (0,91 against 0,59 in the exemplar). It does not say the layout is stable under the assumptions.
- **Stability, declared in advance.** The map is recomputed under the alternative settings written down in step 4 (the exemplar: no correlations, the judgement's correlations, strong correlations) and counts as stable only if every pair of versions agrees in the order of the inter-option distances at or above the threshold (0,8 in the exemplar, Appendix B). Failure is printed, not softened.
- **What the map never claims** (board MV3 § 06): causation; the behaviour of the market or the firm; that a relation is observed rather than a property of the model; that a distance is an effect size in any unit; that the terrain between points is an estimate; validity borrowed from team-communication Sociomaps, which do not cover this use.
- **Internal until approved.** Maps whose observers are futures or properties are an estimate awaiting the methodologist's approval; they print *odhad (metoda čeká na schválení metodikem)* and carry the internal label (R2-P04 § 3.2). The Sociomap of people's ratings is approved since 21. 10. 2026; this is a different mode.

8.4 How the exemplar did it

Eight sites, 20 000 futures as observers, each future centred on the mean of the eight (R2-P04 § 3.3). The map divides the sites by country: the German sites lie together in a valley (74,2 mil. EUR mean saving at Eschfeld), the Czech and Polish sites on a ridge (147,6 at Odrzyca). Within countries the relations are strong (Lindenau–Eschfeld 0,71, Wendelstedt–Eschfeld 0,70; Lipová–Kamenec 0,55; Odrzyca–Bory 0,48); between countries weak and negative, −0,07 to −0,27, the sign partly a consequence of centring (T7). Two of 28 pairs were marked reliable. Split-half agreement 0,91 against a threshold of 0,59: determined. The stability test failed: the judgement version agrees with the no-correlation version at 0,95 and with the strong version at 0,84, but those two agree only at 0,75, under 0,8; the relations in the matrix are steadier than the drawn positions (0,966, 0,899, 0,826). Without correlations seven pairs are reliable and the Czech sites merge; with strong correlations no pair is and Vrbice moves away. The report therefore reads from the map only the heights and the split into a German group and a Czech–Polish side, which appears in every version, and nothing about fine positions, causes or probabilities (R2-P12 § 11.2). For the backup question it uses T8: the best backup fails together with the chosen site in 75–79 % of its bad futures, the German sites in 97–100 %; no site is a backup of another (§ 11.1, 11.3). The version without correlations did not converge in the program's log, one more reason to read positions carefully (R2-P15).

8.5 In AIA today

- The `aia-sociomap-5` engine, signed off by the methodologist on 21. 10. 2026 and what a new run pins: `domain/sociomap/signed_off_inputs.py` (per-person centring, the baseline, pair floors, Benjamini–Hochberg), `signed_off_layout.py`, `signed_off_credibility.py` (split-half agreement, map uncertainty), `signed_off_respondents.py`, `signed_off_segments.py`; `engine_v2.py` `compute_object_map` and the contract-3 artefact.
- The register: which method at which spec fingerprint is approved; anything absent is `INTERNAL_ONLY`, never presumed (`domain/sociomap_register.py`, ADR 0038).
- Map snapshots drawn on the server from the stored map at the report's camera and approved as artefacts (`application/map_snapshots.py`, RD4); the map chapters of the report (`application/report_maps.py`, `infrastructure/report_docx/object_map_figure.py`).
- The canvas's own proof that the engine takes a futures table as rows: `prototypes/obchodni-model/tools/engine_maps.py` ran `compute_object_map` on 2 000 futures and drew the maps with the report's renderer.
- The web's map: `ObjectMapPanel`, `lib/object-map-signed-off.ts`, `lib/sociomap-display.ts`, display only.

8.6 New

- The scenario mode's contract (`sociomapa-systemu#SYS9`): rows a model's draws, the baseline the imposed input correlation, stability as the sensitivity to seeds and correlation settings, no p-value; board MV3 in revision, session D.
- The shared-failure table T8 and the stability test: Spearman on the 28 distances between every two settings ≥ 0,8, three settings declared before computing, a failure printed as the exemplar prints it (`#SYS9b`, **new**).
- Each mode its own Metodika entry, internal until approved (`#SYS12`); the map on Výsledky and in the report (`cost-model-study#CS5`), declared extremes and direction per property (`#CS1`, reused by the weighting map).
- The weighting-scenario map, a second map beside this one, on board MV9 (`sociomapa-systemu#SYS13`, **new**; chapter 11). SI1–SI4 keep only what MV9 reads: states, gates, coverage, the overlap check.

8.7 ChemVitro acceptance

- The eight sites placed by whether they do well in the same futures; links only where the declared rule allows; the view chosen and the reason stated (`STUDIES.md` § 2).
- The map labelled internal until Metodika approves the method.
- The relation matrix and the shared-failure table printed; the backup question answered from the table.
- The stability test declared before the computation, and its result printed whether it passed or failed.
- Criteria: K3 (the map is not part of the recommendation), K6 (the matrix, the text and the figure agree).

CHAPTER 9 · STEPS 8 AND 9

Osnova a Zpráva: the outline, then the report

The outline comes before the text. It lists every claim the report will make in the order of the argument, each with its source or method and why it matters to the decision. The report then says only what the outline holds.

9.1 What goes in

- The results, the map, the input table, the findings table, the public inputs and the decision record: every artefact of steps 1–7, by id.
- The part structure of a type-2 report and the grade and source conventions (below).
- The validity review of chapter 13, before approval.

9.2 What comes out

The outline (*Osnova zprávy*, board MV4)

Five parts in the order of the argument (review R10): 1 the decision and what is asked of the board; 2 why the preferred option is acceptable, or why none is; 3 what could change that conclusion; 4 implementation conditions, owners and dates; 5 supporting evidence and technical appendices. Each point carries its source or method, its grade attributes, and one line "why this matters" (*proč na tom záleží*). A judgement is a typed judgement record; a synthesis from sources is an inference record with its evidence (review I5). Stops (*zastávky*) close parts for review; a closed stop may go to the client's decision-makers for comment. Board MV4 (session E) shows the outline of report 2 as 18 parts with their ids, maps the five parts of the argument onto them (1 = the summary, P01; 2 = chapters 5–6, P06–P07; 3 = chapters 7–8 and 12, P08–P09 and P13, where the weighting-scenario map belongs; 4 = chapter 13, P14; 5 = the rest), prints the seven grades with their four attributes, and opens a number's provenance card: source, method, uncertainty, validation, the run, the inputs it stands on, the data check behind it.

The report (*Zpráva*, board M6)

The exemplar's part structure, which the MVP reproduces:

Part	Content	Rule
R2-P01	*Shrnutí*: *Doporučení* · *Důvod* · *Jistota* · *Další rozhodnutí*, the rule table, *Stav podkladů*	one page plus the state box; the rule table is T1 without the intervals; the state box names the gaps that could change the result and the public inputs not yet verified
R2-P02	*Úvod*: the situation, the options, the rule	the rule printed in full with its date and the interpretation record
R2-P03	*Přehled literatury*	each work with "what follows for this report"
R2-P04	*Metodika*: about AIA, the Sociomap, the model in words, the data, the grades	the grade legend is printed in every report; the Sociomap paragraph says which modes are approved
R2-P05	*Předpoklady*: the assumptions the verdict rests on	the ones without a document first
R2-P06–P11	the result chapters: decision, schedule, levers, information, finance, operations	each chapter's first sentence is its finding; each number carries its grade; sources numbered at the chapter's end
R2-P12	the map chapter	relative and absolute readings, the stability result, what the map may and may not say
R2-P13	*Diskuse*	what the results mean together, and what the model does not resolve
R2-P14	*Doporučení a provedení*	a dated plan: when, what, who, output; the enquiries with their owners; what to watch
R2-P15	*Meze studie*	every judgement without a document, every provisional input, every simplification, every method awaiting approval; "each change computed separately"
R2-P16–P17	*Literatura, Seznam zkratek*	—
R2-P18	*Přílohy* A–D and tables T1–T9	A the document register; B the method in detail and the findings P1; C data checks and computation (seed, N); D the tables, generated by the model from the same run

Every number carries one of the grades of chapter 12 in small type and a numbered source at the chapter's end; the summary's recommendation status says *předběžné* where the recommendation rests on a method outside the register (chapter 12). Board M6 (session E) renders the exemplar's page 1 and chapter 13 as AIA would print them and adds the release state: *Předběžné doporučení*, four unresolved qualifications (sixteen public inputs unverified, the compensation chance a judgement without a document, the investment reading awaiting the CFO, the futures map failed its stability test), one K6 finding, and *Schválit jako předběžné* as the only approval offered.

9.3 Checks

- No point without a source or a method; a point whose source was not captured does not enter the report (MV0 M-9; board MV4's point 3.4).
- The report says only what the closed outline holds. A sentence that compares in words with no number behind it is marked (RD7's sentence check, already built).
- Every number is a reference to a stored result: the same run, the same artefact id. Tables are generated by the model from the run, so they agree with the text by construction (R2-P18 Appendix D; K6).
- The required disclosures are in every report: the recommendation's status and qualifications, each number's source and method, what is outside the model, who stands behind the assumptions, the method versions (board MV4's *Povinná zveřejnění*).
- The validity review is recorded before approval, and the approval records the exact version and the unresolved qualifications (review R1, R9; chapter 13).
- A stop closed with the client's people records their agreement (ADR 0046 W26).
- No respondent, person or e-mail is named beyond a role; the report is Czech in the MVP (ADR 0046 W28).

9.4 How the exemplar did it

The report's first page is the summary in four paragraphs and the rule table; the second is the state of the inputs. Every chapter's title is its finding ("No site yet passes the rule", "The cheapest sites do not meet the deadline certainty", "The saving rests on today's price from Hanshin"). Every number carries its grade in small type, and every chapter ends with *Zdroje a stupně*, numbered. The part ids R2-P01 to R2-P18 let the executive comment part by part. The limits chapter admits what the report cannot claim: the compensation chance is a judgement, public inputs are provisional, the map is not stable, the methods of the futures await approval, each change was computed separately (R2-P15). The implementation plan is a dated table with owners and outputs (R2-P14). The tables T1–T9 are generated from the same run as the text (R2-P18).

9.5 In AIA today

- The report as data: the document model with sections and blocks, numbers only as refs (`domain/report/model.py`, `document_model.py`); the evidence ledger built from admitted claims with print grades (`evidence.py`); the section-source manifest naming each section's sources (`sources.py`); printed numbers as locked content controls with one tag grammar (`control_tags.py`, `controls.py`).
- The DOCX renderer with deterministic bytes, the Word style sheet from the print tokens, tables with repeating headers and suppressed rows counted, evidence marks as vector images (`infrastructure/report_docx/`); the PDF beside it (`infrastructure/report_pdf.py`).
- Composition: the internal report, the client report, the report with literature, their chapters from verified sources (`application/report.py`, `report_chapters.py`, `report_tables.py`, `report_design.py`, `report_method.py`, `client_report.py`, `contextual_report.py`).
- Approval once per version with the approver recorded and self-approval rules (`application/report_approval.py`, `artifact_repository.approval_record`); the client gate's checks as warnings (`domain/report/client_gate.py`, RD0); the sentence check (`sentence_check.py`, RD7); the report editor with saved versions and their changes against what code wrote (`report_editor*.py`, `saved_changes.py`).
- The stored document read on screen with each number a chip to its claim (`ReportDocumentView`, `lib/workbench-artifacts/report.ts`).

9.6 New

- The outline as data with a resolving link on every point (`research-findings-stage#FND0`, `#FND2`), findings in (`#FND3`), tables from points and graphs from tables (`#FND4`, `#FND5`), stops (`#FND8`), *Zpráva* written from the outline (`#FND9`); board MV4 (session E, landed) is the outline of report 2 itself.
- Composition per situation from chapter kinds and outline presets (ADR 0045; `situation-reports#RC1`), fixed method texts signed by the methodology owner (`#RC4`); the report-2 recipe: R2-P01…P18 as an outline preset with part ids, the grade legend, numbered sources per chapter, *Stav podkladů* and *Literatura* (`research-findings-stage#FND13`, **new**); the argument order (review R10).
- The frame and CALCULATED (`cost-model-study#CP8`), ESTIMATED (`#CP10`), the method appendix (`#CR9`), and the seven print grades mapped to the code's with an admission rule each (`#CP8b`, **new**; chapter 12).
- Approve and export (`report-delivery#RD3`); tables generated from the run into the report (the exemplar's T1–T9, `#FND4`); the limits chapter composed from the input table's states and the run's declared simplifications (part of `#FND13`).

- The validity review before approval: the five questions, extreme-input tests and a named reviewer recorded on the ledger (`research-findings-stage#FND14`, **new**; board MV8, chapter 13); the release label of R9 where the owner decides it (board M6, *Stav vydání*).
- The state box and the limits chapter composed from the input table: board MV4's P01 takes its count from T9 (16 of 67) and names the text's 54 as the K6 finding; the *prozatímní* banner on every page while one decisive input is provisional (board MV2 § 02b).

9.7 ChemVitro acceptance

- Every number carries its grade and source: file and row or page, or the quoted public source (`CASE.md` § 11).
- An estimate from a method that is not approved is labelled *odhad (metoda čeká na schválení metodikem)*.
- The report is Czech, as PDF or DOCX; the seven parts of `STUDIES.md` "What every report contains" are present.
- No respondent named or quoted by id.
- Criteria: K6 is checked here, by code and by a person: no contradiction between narrative, tables and figures, percentile and probability wording included. The canvas's own K6 finding (SI3 against KM4) shows the check is needed.

CHAPTER 10

Built vs designed

No type-2 code is on `develop`. The exemplar's model ran outside the product. What AIA has is the ground the pivot builds on: the shell, the working copy, client sources, Deep Research, the Sociomap engine and its register, the report document model and its renderer, approvals and audit. This chapter says which is which.

10.1 The state on 8 November 2026

Step	Built on `develop` (reused)	Designed (board; plan chunk as the scope names it; **bold** = new, in no plan file yet)	Neither yet
Zadání	working copy, revisions, edit ops; Brief stage; the Lead; approval ledger; material classes	Z1–Z7; `decision-brief#DB0`, DB2–DB5, **DB3b**; `cost-model-study#CP3`	the perspectives chosen per case
Data	client sources with readers and locators; dataset preview with personal columns; intake sheet; evidence policy (ADR 0027/0037)	M1, MV1; `client-data-in-analysis#2`, 2a, 3, 3a, 3b, 5, 10, 11, 12, **13**; `cost-model-study#CR1`	—
Rešerše	Deep Research with grounding and the sealed bundle; RetrievalGate; Wikipedia, Brave, Common Crawl; verified sources	MV2, MV2b; `deep-research-v2#DRV1`–`DRV3`, `DRV5`; `public-data-routes#PDR1`, `PDR2`; `legal-context#LGL1`–`LGL3`	verification at source as a dated act, named in `DRV1`'s state
Předpoklady	methodology as data (versions, activation, bounds); ledger and audit; the simulation core's typed, validated inputs as a pattern	M2; `cost-model-study#CP1`, `CR2`, `CR3`, **CR3b**, **CR1b**; `client-data-in-analysis#12`	mixtures for expert disagreement (board M2 § 02)
Model	worker, workflows, artefacts with SHA-256, pipeline fingerprints; methodology pins on a run; seeded determinism	M3, M4; `cost-model-study#CP2`, `CP3`, `CP5`, `CP6`, `CM7`, **CM8**, `CR3`, `CP10`; `quantitative-families#FIN1`, `FIN2`	—
Výsledky	admitted claims and suppression; results reading (unknown never ordered); print grades; table exports	M4, M4b, M4d, new M4g; `cost-model-study#CR7`, **CR7b**, `CM2`, `CR4`, `CR6`, **CR6b**, `CR8`, `CP8`, **CM9**; `quantitative-families#FIN3`; `legal-context#LGL3`	—
Sociomapa	`aia-sociomap-5` engine; the register; map snapshots; report map chapters; the canvas tool that maps futures with the engine	MV3, SI1–SI4, new MV9; `sociomapa-systemu#SYS9`, **SYS9b**, `SYS12`, **SYS13**; `cost-model-study#CS1`, `CS5`	the register entries themselves (the methodologist's)
Osnova, Zpráva	document model, evidence ledger, control tags, source manifest; DOCX and PDF renderers; composition; approval; client gate; sentence check; report editor	MV4, M6, MV8, MV5; `research-findings-stage#FND0`, `FND2`–`FND5`, `FND8`, `FND9`, **FND13**, **FND14**; `situation-reports#RC1`, `RC4`; `cost-model-study#CP8`, **CP8b**, `CP10`, `CR9`; `report-delivery#RD3`	the release label of R9 (the owner's)

10.2 The plans that carry the build

From `python tools/progress.py` on 8 November, on `develop`: `cost-model-study` is planned, 0 of 32 chunks; `client-data-in-analysis` in progress, 1 of 15 (8a built); `report-delivery` in progress, 7 of 16; `quantitative-families` and `situation-reports` planned. The other plans the scope cites (`decision-brief`, `deep-research-v2`, `public-data-routes`, `legal-context`, `sociomapa-systemu`, `research-findings-stage`, `analysis-frameworks`, `diagrams`, `register-personal-data`) exist only on branch `chore/pivot-business-decisions`, with ADR 0046, Proposed and not merged; the chunks the scope marks **new** exist in no plan file at all. ADR 0044 (quantitative models as a study kind) and ADR 0045 (reports composed per situation) are Proposed on `develop`. The scope (§ 2, § 7) maps each step to its chunks and orders them in five waves; this chapter's "Designed" column is that mapping.

10.3 Three honest statements

- **The exemplar was not produced by AIA's product.** Its author computed the model with his own program and wrote the report by hand to AIA's conventions. The report's sentence "every number was computed by a verifiable program" is true of that program. The MVP's job is to make it true of AIA.
- **The canvas is design.** The thirty boards show the screens and the method; their numbers are fictional, and the Velis case's numbers are illustrative, not validation evidence (board MV5). The boards hold the method; the plan chunks hold the build.
- **The methods of the futures are not registered.** Monte Carlo with PERT and correlations, the decision boundary, both measures of the value of information, the futures map and the weighting-scenario map await the methodologist's entries (chapter 13). Until then their results print *odhad (metoda čeká na schválení metodikem)*, and a recommendation that rests on them is *předběžné* under R9, which itself waits for the owner.

CHAPTER 11

Perspectives, components and the two maps

The owner asked for every business perspective: financial, strategic, operational, marketing, logistics, "just everything". A perspective is a set of components of the decision; a component is one measure per option with a source, a grade and a direction. The components feed one new map, the weighting-scenario map, which sits beside the futures map and is never merged with it. This chapter takes its definitions and ChemVitro's components from the scope, § 3 and § 4.

11.1 Perspectives and components

A **component** is one measure per option, with a **source** (a model result, a client-data number, a cited finding, or a named expert's score), a **grade** (one of the exemplar's seven, chapter 12) and a **direction** (↑ higher is better, ↓ lower is better). A **perspective** is a set of components. The weighting-scenario map scores components; the client's rule does not read them.

Five rules, from the review of 5 November:

- **Gates are not components** (R2). The rule's four conditions are pass / fail / unknown per option. An unknown gate is "cannot be recommended until evidenced", and no weight compensates a failed one.
- **A missing component keeps a range, never a score** (R2). The cell shows the range it could take; the map is drawn at the documented minimum with the range beside it.
- **One mechanism, one component** (R3, "no double counting"). A component whose mechanism the model's saving already holds is marked *v modelu*; one the model does not hold is *mimo model*. A scenario may weight the saving or its parts; the overlap check (§ 11.3, line 6) refuses a vector that weights both.
- **Judgement is labelled** (A3). A component scored by a person carries that person's name and the grade *úsudek*.
- **An empty perspective stays and says so.** For ChemVitro the marketing perspective is empty; for Velis it is central.

ChemVitro's components

Sources are the exemplar's tables and parts. *Stejné pro všechny lokality* names a build-versus-buy component that does not separate the sites; it is shown in the report and not scored on the map. The last column says whether the saving already holds the component's mechanism.

Perspective	Component (per site)	Source	Grade	Direction	In the saving?
Finanční	*Medián úspory proti nákupu, NPV 2027–2044 při 8 %*	T1, R2-P06	odhad	↑	the saving itself
	P(porazí nákup) ≥ 0,75	T1	odhad	gate	
	P(ztráta > 20 mil. EUR) ≤ 0,10	T1	odhad	gate	
	Investice v základním případě ≤ 175 mil. EUR	T5	výpočet	gate; ↓	*v modelu*
	P(investice ≤ 175)	T5	odhad	↑	*v modelu*
	IRR; rok návratnosti	T4	výpočet	↑; ↓	*v modelu*
	NPV po zdanění při 8 %	T4	výpočet	↑	*v modelu* (tax is a fixed parameter, I4)
	Forma pobídky: daňová úleva (CZ, PL) nebo dotace (DE), podíl a P(schválení) 0,5	R2-P10, T9	z dat klienta; právní předpis; úsudek	informational	*v modelu* (the after-tax view)
Strategická	*Společné selhání s vedoucí lokalitou: podíl budoucností, v nichž je i tato lokalita horší než nákup*	T8, R2-P12	odhad	↓	*mimo model*
	Závislost na jednom dodavateli; přirážka při prodloužení 12–22 %; podmínky po roce 2035; zůstatková hodnota 30 %	R2-P03 § 2.4, R2-P08 § 7.2, T9	z dat klienta; úsudek	*stejné pro všechny lokality*	levers of build vs buy
	Tržní pozice	—	—	not applicable: the plant supplies Brno (62 %) and Pardubice (38 %) only	
Provozní	*Provozní náklad na tunu 2032, celkem*	T6	výpočet	↓	*v modelu*
	z toho energie; údržba a park; odpad a voda	T6	výpočet	↓	*v modelu*
	Najíždění 60–90 % v prvním půlroce	T9	úsudek	*stejné pro všechny lokality*	

Perspective	Component (per site)	Source	Grade	Direction	In the saving?
Marketingová / obchodní	—	—	—	empty for ChemVitro: captive demand, no customer promise at stake. Velis: customer promise, delivery time, demand growth	
Logistická	*Doprava produktu do Brna a Pardubic, EUR/t*	T6	výpočet	↓	*v modelu*
	Vlečka k postavení: délka, cena za km, získání pozemků (Odrzyca)	T5, T9	z dat klienta; odhad; úsudek	↓	*v modelu* (capex, delay)
	Silniční přístup: nosnost mostu 16 t proti 40 t, zesílení (Eschfeld)	T5, T9	z dat klienta; úsudek	↓	*v modelu* (capex, start)
	Doprava surovin přes překladiště do stavby vlečky, EUR/t	T9	úsudek	↓	*v modelu*
Prostředí a povolování	*Stanice nad limitem memoranda (BaP, PM2,5) → šance kompenzace 0,8 / 0,2 / 0*	R2-P05, T9	z dat klienta; úsudek	↓	*v modelu* (the decisive input)
	Prodloužení povolování s kompenzací, měsíce (CZ 9–15, PL 12–18)	T9	z dat klienta	↓	*v modelu*
	P(výroba k 1. 7. 2030) ≥ 0,80	T1, R2-P07	odhad	gate	
	Předčištění odpadních vod: potřeba a cena (Lipová, Lindenau)	T5, T9	z dat klienta; odhad	↓	*v modelu*
	Poplatek chemického parku (Wendelstedt)	T9	z dat klienta	↓	*v modelu*
	Emise CO_2 a EU ETS, 56 EUR/t	T6	výpočet	*stejné pro všechny lokality*	
Lidé	*Náklady na zaměstnance na tunu (mzda operátora × skladba mezd × odvody)*	T6, T9	výpočet; právní předpis (*prozatímní*)	↓	*v modelu*
	Doba obsazení místa operátora, dny (Eschfeld 118, jinde 49–71)	R2-P07 § 6.2, *Trh práce v regionech*	z dat klienta	↓	*mimo model* (stress test only)
	Velký nábor v okolí (Kamenec: 380 lidí do 2028)	*Trh práce v regionech*	z dat klienta	↓	*mimo model*
	Růst mezd po roce 2031	T9	úsudek	*stejné pro všechny lokality*	

What the table shows for the map: 22 scored components, of which 18 are *v modelu*. A scenario that weights "finance 0,5 · provoz 0,3 · logistika 0,2" over these scores operations and logistics twice, because the saving already holds their cost per tonne. Two honest scenario families exist: **(a) the saving against what the model leaves out** (shared failure, hiring time, hiring competition), and **(b) the parts of the saving against each other** (people, maintenance, transport, permitting cost) without the saving itself. The map records which family a scenario belongs to. Which family the owner's example means is his open decision (chapter 13).

11.2 The futures map

Chapter 8 describes it. In one line each: rows are the model's draws; the claim is about the model; the cell is the rule's measure; centring per future makes the reading relative; the uncentred shared-failure table reads the absolute; links only where the mode contract allows; a stability test over settings declared in advance; relative apart from absolute; internal until the futures mode has its register entry.

11.3 The weighting-scenario map (*Mapa scénářů vah*, board MV9)

The owner's request: "allocate different weights to different components … whether logistics is a 0,3 and operations management is 0,5 … run the scenarios and display the scenarios on the Sociomaps." The ten lines of its contract, as the scope writes them (§ 4; chunk `sociomapa-systemu♯SYS13`, **new**):

1. **Objects** are the options: the eight sites. The buy baseline is the rule's reference, not an object.
2. **Rows** are named weight vectors over perspectives or components, composed by the researcher or the client's people ("FR: finance 0,5 · provoz 0,3 · logistika 0,2"), each with who composed it and when, recorded in the approval ledger before results, by a composer who has not seen them (swing weighting, R3).
3. **The fan** around each row is k perturbed vectors from a declared concentration; it is a sensitivity setting, never a probability (R4). The map prints *podíl testovaných nastavení, v nichž je možnost první*, never P.
4. **A cell** is the option's weighted score: each component scaled on its declared extremes and direction (`cost-model-study♯CS1`), gates excluded and shown as pass / fail / unknown beside the map (R2).
5. **A missing component** keeps its range: the cell is an interval, the map is drawn at the documented minimum, and the row says what the top of the range would change (R2).

6. **The overlap check** marks components with one mechanism (§ 11.1, *v modelu*); a vector that weights the saving and one of its parts is refused by name and offered the two honest families.
7. **Height** is the weighted score under the scenario the reader chooses; relative (centred per row) and absolute readings side by side, as MV3's contract for the scenario mode (R5).
8. **Layout** is the `aia-sociomap-5` engine's over the rows × objects table, as the prototype already runs it (`tools/engine_maps.py`, `compute_object_map`); its pair statistics are not read, because the rows are settings we chose (MV3). Options close together win under the same weightings.
9. **Apart from the futures map.** The futures map's rows are the model's draws; this map's rows are preferences. They sit side by side and are never merged.
10. **Where it goes.** The map is preference information. It enters the report under *Co by závěr změnilo*, never the recommendation; the client's rule decides (R3). Its method is a Metodika register entry of its own (`sociomapa-systemu#SYS12`), internal until the methodologist approves it, labelled *metoda čeká na schválení metodikem* wherever it appears.

What the exemplar shows for it

The exemplar has no weighting map; it has the material for one. Its components per site are in T1, T4, T5, T6, T8 and T9, and its text already weighs them in words: the Czech and Polish sites win on finance and operations, Wendelstedt on the schedule, the German sites on permitting certainty, and none on all. A family (a) map would show that the three leaders win under every weighting that keeps the saving first, and that only a heavy weight on shared failure or hiring competition moves Kamenec and Odrzyca apart; a family (b) map would show where the German sites lose (people, maintenance, transport) and where they do not (permitting cost). Wendelstedt's deadline certainty is a gate, not a weight, and the gate is what keeps it out. That is the kind of statement the map may make: what the preferences would favour, not what the rule chooses.

> **In revision.** The definitions and the component table above are the scope's (session A, 8 November). Session D builds MV9 on the real engine from them; the fan's k and concentration and the scenarios' names wait for D's board and for the owner's and the methodologist's answers (chapter 13).

CHAPTER 12

Grades, sources and the release gate

Every number in a type-2 report carries a grade in small type and a numbered source at the end of its chapter. The grade says how the number came to be; the source says where. Behind the one printed grade stand four attributes.

12.1 The exemplar's seven grades

Grade	Meaning (R2-P04 § 3.5)	Exemplar instance
z dat klienta	a number read from a client's file or document	contract end 30. 6. 2030; 36 comparable plants
výpočet	a number the program computed from stated inputs in one computation: the base case, break-even, finance, operations	base-case saving 170,7 mil. EUR; T4–T6
odhad	a range derived from client data, for instance from comparables; and every number from the futures, which carries the full label *odhad (metoda čeká na schválení metodikem)* while the method is unregistered	overrun median 17 %; every median, interval, probability, value of information, boundary, delay table and the futures map
úsudek	the author's assumption where the client has no answer; also proposed dates	the 20 % compensation chance; the correlations; the plan's dates
veřejný zdroj	a figure from public research, provisional until verified at the source	allowance price 89,10 EUR/t; exchange rates
právní předpis	a provision, likewise provisional until verified	tax rates, employer contributions, the directive's limits
průzkum	answers from a survey; type 2 does not use it	—

12.2 The four attributes behind a grade

Version 2 of this methodology, following the review's I3, split a number's label into four attributes. The printed grade is a projection of them; the outline and the screens hold all four (board MV4 prints the seven grades with their attributes and where each occurs in report 2; the result boards' chips show the printed grade).

Attribute	Values	Projection to the printed grade
Source	*data klienta · veřejný zdroj · právní předpis · úsudek · model*	the source alone gives *z dat klienta*, *veřejný zdroj*, *právní předpis*, *úsudek*
Method	*přímo · výpočet · odhad · modelováno*	a model source prints its method: *výpočet* or *odhad*; *modelováno* is the panel respondents' grade (ADR 0024) and does not occur in type 2
Uncertainty	a range, an interval, a probability's basis, or a certainty	printed beside the number, not in the grade: "90 % interval −23,8 to 359,1"
Validation	*metoda v registru · čeká na metodika*; for a public input *ověřeno · prozatímní*	adds the suffix *(metoda čeká na schválení metodikem)* to *odhad*, and the state *prozatímní* to a public input

In the code the print grades are five today, `PrintGrade` in `domain/report/evidence.py`: measured, calibrated, modelled, holdout-pending, unknown, strongest first, a claim printing the weakest grade of the fields behind it, a missing grade printing "?". The plan adds CALCULATED (*výpočet*) and ESTIMATED (*odhad*) with the register's status (`cost-model-study#CP8`, `#CP10`), and ADR 0027 the client-reported origin. The scope's `cost-model-study#CP8b` (**new**) maps all seven in one table with an admission rule each: *z dat klienta* = CLIENT_REPORTED, *výpočet* = CALCULATED, *odhad* = ESTIMATED, *úsudek* = a labelled judgement with its author, *veřejný zdroj*, *právní předpis* (`legal-context#LGL5`), *průzkum* = the survey origins. The provisional state is `deep-research-v2#DRV1`'s. The weakest-grade rule carries over: a number computed from a judgement input prints *výpočet* but its sensitivity to that input is shown; a number from the futures prints *odhad* whatever its inputs.

12.3 Sources, numbered per chapter

- Each chapter ends with *Zdroje a stupně*: a numbered list; each printed number cites a list entry by its number (R2-P02: "z dat klienta 1").
- An entry names the document by its cited name and page or paragraph ("Zápis zadání s klientem §6", "Dopis dodavatele, s. 1"), the file for data, the table for a computed value ("výpočet: tabulka T5 v příloze"), or the method's status for an estimate ("odhad (metoda čeká na schválení metodikem)").

- An entry may carry two grades when a number rests on both ("výpočet; z dat klienta: Srovnatelné závody"); the weaker is printed at the number.
- The document register (Appendix A) maps every cited name to its file, so a source is checkable with one open.
- In AIA, the section-source manifest (`domain/report/sources.py`) already records each section's sources by artefact id and SHA-256; the numbered list is its printed form.

12.4 The provisional state

Prozatímní is a validation status, not a grade. It marks a public input not yet verified at the source, and it is printed in three places: the input table's state column (T9), the sentence that uses the input ("dlouhodobý vývoj cen energie (prozatímní údaj)"), and the first-page state box with the count (16 of 54). It is removed only by a dated verification by a named person (board MV2 § 02b). The report's running head carries it while any decisive input is provisional: the exemplar prints *Pracovní verze · prozatímní* on every page. The count itself is a claim the check reads: the exemplar's text says 16 of 54, its T9 has 67 rows, and boards MV4, M6 and MV8 record that as the report's one K6 finding and take the count from the table.

12.5 The release gate (R9, proposed)

> **Waits for the owner.** R9 changes two earlier decisions: W27 (an estimate whose method is not registered still reaches the client report, labelled, with a warning, never refused) and report-delivery RD0 of 22 October ("warnings for all of it"). Boards MV0 (M-5), MV5 (§ 04) and M6 show it as proposed.

As proposed, for a business-decision report:

- Every decision-critical claim is checked twice: support in its source (code) and correct interpretation (a person who is not the claim's author).
- A recommendation that rests on a method outside the register is *předběžné doporučení* and is never released as approved investment advice. The exemplar's recommendation rests on the futures, so it is provisional in this sense, and its summary would say so under R9.
- Exploratory material enters the report only with explicit limits ("exploratory; does not support the recommendation"). The weighting-scenario map is such material.
- Approval records the exact version and the unresolved qualifications; "approved" means ready in substance, not merely checked.
- The Deep Research thresholds of M-7 (citation precision ≥ 90 %, unsupported claims < 10 %) are one check, not the release criterion.

What exists today: approval once per stored version with the approver and self-approval rules recorded (`application/report_approval.py`); the client gate's checks as warnings recorded on the approval's ledger row (`domain/report/client_gate.py`, RD0); the sentence check (RD7). What R9 adds: the recommendation's status as a field of the approval, the qualifications list, and the second, human reading of each decision-critical claim.

CHAPTER 13

Governance

Reproducibility proves the code computes what is written. It does not prove the model is the right one for the decision. That needs a review by someone who did not accept the inputs, a register that says which methods are approved, and a record of what is still open and whose word closes it.

13.1 The validity review (R1)

Before a recommendation is approved, an independent reviewer answers five questions; the researcher who accepted the inputs answers to them, never reviews them (board MV8 § 01; review R1).

#	Question	In the ChemVitro exemplar
1	Is every option represented comparably?	the same plant, horizon, rate and timeline; site-specific months and costs from the site's documents (R2-P02, Appendix B)
2	Are the material costs, benefits, constraints and implementation dependencies in the model?	the EPC exclusions priced; outside the model: transfer prices, hiring time in the schedule, raw-material freight, the annual step (R2-P15)
3	Could an excluded item reverse the recommendation?	raw-material freight would lower savings by 9–14 mil. EUR without changing the verdict (P1); the 2030 air limits could (R2-P01)
4	Do extreme inputs give sensible results?	seven of the report's own computations read as extreme-input tests on board MV8 § 03: compensation chance 0 % and 30 %, the 2025 price, the post-2035 premium at 0 % and 30 %, the investment read "in every future", a six-month delay without compensation; all seven behave as expected
5	Has an independent reviewer challenged the structure and the decisive assumptions?	not recorded in the exemplar; board MV8 (session E) records a reviewer who did not work on the study, assigned 5. 11., with three questions open: the compensation chance, the post-2035 terms, the investment reading

With the questions, board MV8 holds a table of the ten limits the exemplar admits in its chapter 14, each with the check that catches it and its state: three caught (the map's failed stability test, the 91 kt outlier, the simplifications listed with their effect), seven open (the compensation judgement awaiting the authorities' document, the stations' distance from the sites, the sixteen provisional inputs, the 2030 directive, the version without correlations that did not converge, combinations not computed, the investment reading awaiting the CFO). Then the statement that validity is not reproducibility and the six criteria K1–K6 with their state on report 2: K1 open (two judgements without a document), K2 partial (combinations and convergence), K3 met, K4 waiting for the recomputation after verification, K5 met, K6 one finding. The review's record is part of the approval; its open items are the four qualifications that keep the recommendation *předběžné*.

13.2 Register entries the exemplar waits on

The Sociomap register (`domain/sociomap_register.py`) and Metodika's versions (`domain/methodology/`, ADR 0023) hold what is approved. A type-2 study needs entries that do not exist. Until each exists, its results print *odhad (metoda čeká na schválení metodikem)*. Board MV5 (session E) carries the same list as the register's rows for report 2, with what the methodologist signs for each, and keeps NPV, IRR, payback, break-even and the tornado as *v registru*: a single computation needs no entry.

Method	What the entry must fix	Where it is used
Monte Carlo over the input table: PERT, declared correlations (Iman–Conover), empirical joint draws from comparables, yes/no events, mixtures	the default distribution by source; the correlation method; the stopping rule and its margin; the seed convention	every median, interval and probability (T1–T3, T5)
The decision boundary	the grid over an input's range; recomputation of the futures per value; how the flip is reported	R2-P06 § 5.3
Value of information for the rule, and EVPPI	the partition of futures by the answer; the rule applied per partition; EVPPI by Strong and Oakley (2013); the order of the two measures; the enquiry table's columns	R2-P09
The futures map (observers = futures) under `aia-sociomap-5`	the mode contract: which pair statistic, if any; the determinedness test; the stability test's settings, agreement measure and threshold; the relative and absolute readings (`sociomapa-systemu#SYS9`, `SYS9b`, `SYS12`)	R2-P12, T7, T8

Method	What the entry must fix	Where it is used
The weighting-scenario map	states, gates, swing weights, value scales, the fan's concentration and k, shares of tested settings, the overlap check (`sociomapa-systemu#SYS13`, `SYS12`)	board MV9
Estimates from comparables	the filter, the statistic, the minimum sample, how a joint draw is recorded (`cost-model-study#CR3b`; `client-data-in-analysis#12`, an estimate from client rows as a Metodika method)	overrun and delay, pre-treatment, spur costs (T9)

13.3 The methodologist's open questions

ADR 0046 lists ten (§ Open: multiple testing on maps; levels or changes; effective sample size; resampling for 24–36 rows; the surrogate null; the relation statistic; the synthetic suite's thresholds; the discount rate in probabilistic rules; PERT inputs; forecast evaluation), with board MV5's recommendation for each. The exemplar and the scope (§ 8) add eight; board MV5 § 02 asks six of them with a recommendation each (PERT inputs; input correlations by Iman–Conover, a Gaussian copula or the common-plant draw; the stopping rule; the discount rate; whether pair statistics are read on the futures map; the stability threshold) and defers the unit and period modes' questions:

1. **The futures map's contract** (R5, `sociomapa-systemu#SYS9`): rows are 20 000 chosen draws, so what does it print instead of the respondent method's "2 of 28 reliable" and split-half 0,91 against 0,59 (R2-P12 § 11.2)? The exemplar drew links by the `-5` pair test with the false-discovery correction; board MV3's scenario contract says no null applies when the model creates the relations.
2. **The stability test** (R2-P18 B): Spearman on the 28 distances between every two of three correlation settings, threshold 0,8; approve, or replace the statistic and the threshold. The "strong" setting (0,95 / 0,85 / 0,4) is the author's judgement.
3. **The empirical joint draw** (`cost-model-study#CR3b`): one comparable plant per future carrying overrun and delay together, as an alternative to PERT with a declared correlation; whether 36 plants support it and how the 91 kt outlier is treated.
4. **PERT λ, the number of futures and the convergence rule** tied to the rule (±0,5 pp on each pass probability, R6); and when a mixture or a uniform is the better default for the 22 inputs the files do not hold.
5. **Value of information for the rule** (`cost-model-study#CR6b`, R7): the rule re-applied per answer branch; approve the definition, including its meaning when the rule's choice is "nothing" and how it combines across several enquiries (the exemplar's 152,1 for all three authorities).
6. **The weighting map's fan**: the concentration and k, and the statement that its shares are shares of tested settings (R4).
7. **The seven print grades mapped to the code's** (`cost-model-study#CP8b`, Q1): one table, Czech labels, admission rules.
8. **A register entry** for `aia-sociomap-5` with futures or settings as observers (today's entry covers respondents' ratings), or a separate method id per mode (`sociomapa-systemu#SYS12`).

13.4 The owner's open decisions

Id	Decision	Where it stands
R9 / Q7	The release label: a recommendation resting on a method outside the register stays *předběžné doporučení* (MV0 M-5); it changes W27 and RD0 of 22 October. The exemplar labels every such number *odhad (metoda čeká na schválení metodikem)* and still recommends; which of the two does report 2 ship with?	the canvas recommends accepting R9 for the business study's recommendation only, warnings elsewhere staying (MV0 § 06); the scope asks the same (§ 8, 1)
Scope 2	The weight vectors: is "finance 0,5 · provoz 0,3 · logistika 0,2" meant as family (b) of chapter 11 (the parts of the saving against each other), or as family (a) (the saving against what the model leaves out)? The overlap check refuses the mix; the owner's example needs one reading.	open
Scope 3	Who may compose a weighting scenario: the researcher alone, or the client's people through the shared review (`research-findings-stage#FND12`, parked)?	open
Scope 4	The Velis boards M4c and M4e are marked parked; keep them on the canvas as parked, or move them off the path?	open
Q1	The four attributes of a number as the vocabulary of the outline and the report	the legend on M6 and the labels on MV4; the mapping table is `CP8b`
Q2	Enforce data classes before a client's data flows into a model	yes for Class A in staging and production, recommended
Q4	The SOMECS name and licence	until checked, the map is *Sociomapa systémů*

Settled by the brief of 8 November: the case for the review's step 5 is ChemVitro, and the MVP's spine is type 2 end to end on it. Settled by the scope: the type-2 model is a model type of its own, `site_cost_of_supply` (`cost-model-study#CM8`), not an extension of *scénáře nákladů*. Settled by the exemplar's client, not the owner: the interpretation of the investment condition (the CFO confirms).

13.5 Change propagation (K4)

A revised input is a new revision of the input table. Every result, table, map and approval that read the old revision is marked stale until someone reviews it; the report's version carries the revision it was composed from. In AIA today the pipeline's fingerprints and the impact rule (`domain/pipeline.py`, the research impact route) do this for a research design; the model platform reuses the rule (`cost-model-study` CP5: inputs frozen by artefact id and SHA-256).

APPENDIX A

Glossary

Term	Meaning in this methodology
Base case (*základní případ*)	one computation with every input at its likely value; compares options and feeds sensitivity; not an estimate of the saving
Baseline	the option every other is compared with; for ChemVitro, keep buying
Component	one measure per option with a source, a direction, a value scale and a state; perspectives group components
Data contract (*datová smlouva*)	which columns of a client file are used and how; units, currency, price year, personal columns, period; accepted by a person
Decision boundary (*hranice rozhodnutí*)	what the rule chooses as one assumption moves over its range, the futures recomputed for each value
Disposition	what was done with a data finding: used; corrected with the reason; left out; taken as an assumption with a range
EVPPI	expected value of partial perfect information: the gain in mean outcome from knowing one input before deciding (Strong and Oakley 2013)
Fan	k perturbed weight vectors around a named weighting scenario, from a declared concentration; a sensitivity setting, not a probability
Futures (*budoucnosti*)	the seeded draws of all inputs at once; probabilities are shares of them
Futures map (*mapa budoucností*)	a Sociomap whose observers are the futures and whose objects are the options; centred per future
Gate	a non-compensable requirement; fail or unknown blocks the recommendation
Grade (*stupeň*)	the small-type label on a printed number: how it came to be; chapter 12
Guarantor (*stojí za ním*)	the person who stands behind an input's range
Interpretation record	a reading of a rule's condition that the rule does not fix, with the alternative, its effect and who confirms
Mode contract	per Sociomap mode: what a row is, what the claim is about, what is measured, what the null means, the resampling, what the stability measures
Overlap check	the refusal of a weight vector that weights the saving and one of its parts; it offers the two honest families instead
Outline (*Osnova zprávy*)	the report's claims in the order of the argument, each with its source and why it matters, before any text
Perspective	a set of components: financial, strategic, operational, marketing/commercial, logistics, environment and permitting, people
Provisional (*prozatímní*)	a public input not yet verified at the source
Register (*registr metod*)	the methods the methodologist has approved, each at its version and fingerprint
Rule (*pravidlo*)	the client's decision rule, written before any result: the measure, the choice, the conditions with thresholds
Shared failure	when one option is worse than the baseline, the share of those futures in which another is worse too (T8)
Share of tested settings	the share of a fan's weight vectors under which an option ranks first; not a probability
Stability test	the map recomputed under settings declared in advance; stable only if the distance orders agree above the threshold
Stopping rule	how many futures to run: until each condition's verdict is known within a margin, or a declared N
V modelu / *mimo model*	whether the saving already holds a component's mechanism; a scenario of family (a) weights the saving against components *mimo model*, one of family (b) the parts of the saving against each other
Value for the rule (*hodnota pro pravidlo*)	what knowing an answer in advance and applying the rule again would bring, averaged over the possible answers
Weighting-scenario map (*mapa scénářů vah*)	a Sociomap whose rows are weight vectors and whose objects are the options; height is the weighted score; preference information

APPENDIX B

The exemplar in numbers

Condensed from the exemplar's tables (R2-P18). Every value is the exemplar's; the grades are its grades. NPV 2027–2044, mil. EUR, pre-tax, 8 % unless stated. odhad marks a number from the 20 000 futures (method awaiting approval); výpočet a base-case computation.

B.1 T1 · The rule at 8 %

Site	Median saving	90 % interval	P(beats) ≥ 0,75	P(loss > 20) ≤ 0,10	P(start by 1. 7. 2030) ≥ 0,80	Investment ≤ 175	Passes
Vrbice (CZ1)	110,8	−40,7 … 331,5	0,88 ✔	0,08 ✔	0,21 ✘	160,6 ✔	no
Lipová (CZ2)	126,0	−30,2 … 349,4	0,91 ✔	0,06 ✔	0,76 ✘	158,6 ✔	no
Kamenec (CZ3)	132,4	−23,5 … 356,9	0,92 ✔	0,05 ✔	0,76 ✘	152,3 ✔	no
Wendelstedt (DE1)	82,3	−74,7 … 308,1	0,80 ✔	0,15 ✘	0,94 ✔	171,8 ✔	no
Lindenau (DE2)	86,5	−68,8 … 311,7	0,81 ✔	0,14 ✘	0,77 ✘	170,8 ✔	no
Eschfeld (DE3)	61,2	−94,4 … 285,7	0,73 ✘	0,21 ✘	0,67 ✘	175,5 ✘	no
Odrzyca (PL1)	134,7	−23,8 … 359,1	0,92 ✔	0,05 ✔	0,75 ✘	148,9 ✔	no
Bory (PL2)	123,2	−30,9 … 344,4	0,91 ✔	0,06 ✔	0,19 ✘	143,9 ✔	no

Median, interval and probabilities odhad; investment in the base case výpočet. At 6 % and 10 % the verdict is the same (T2, T3): the deadline probability does not depend on the rate. Keep buying costs 897 mil. EUR in the median of the futures and 911,6 in the base case.

B.2 T4 · Finance, base case

Site	Pre-tax 6 %	8 %	10 %	Post-tax 8 %	Median post-tax 8 %	IRR	Payback	Start (base)
Vrbice	189,2	134,5	92,7	96,6	87,5	18,2 %	2035	2030,92
Lipová	219,3	159,9	113,8	116,3	99,9	19,2 %	2034	2029,92
Kamenec	225,6	165,9	119,6	121,3	107,3	19,9 %	2034	2029,92
Wendelstedt	149,8	99,8	61,2	84,3	72,1	14,8 %	2035	2029,92
Lindenau	156,0	105,7	67,0	92,4	80,0	15,5 %	2035	2030,17
Eschfeld	139,4	91,2	54,2	69,3	45,9	14,4 %	2036	2030,25
Odrzyca	230,6	170,7	124,2	118,6	107,2	20,5 %	2034	2029,92
Bory	200,5	145,8	103,9	99,7	96,0	20,1 %	2035	2031,17

Base case výpočet; the post-tax median odhad. Tax rates provisional. Incentives are not in the base case; the futures grant them with probability 0,5.

B.3 T6 · Operating cost per tonne in 2032, base case, EUR/t

Site	CO_2	Freight	Energy	People	Waste	Raw materials	Water	Maintenance, park	Total
Vrbice	56	38	179	121	17	760	26	113	1 311
Lipová	56	20	179	116	20	760	25	124	1 300
Kamenec	56	24	179	124	16	760	26	111	1 295
Wendelstedt	56	58	182	203	13	760	25	157	1 455
Lindenau	56	53	190	195	15	760	32	132	1 434
Eschfeld	56	55	190	232	14	760	34	122	1 463
Odrzyca	56	37	192	102	18	760	21	105	1 292
Bory	56	33	192	109	17	760	20	103	1 290

výpočet. A bought tonne would cost about 2 340 EUR in 2032 in the base case, premium and freight included.

B.4 T7 · Relations between sites across the futures

	Vrb	Lip	Kam	Wen	Lin	Esch	Odr	Bory
Vrbice	–	0,50	0,52	−0,27	−0,27	−0,26	−0,15	−0,07
Lipová	0,50	–	0,55	−0,22	−0,23	−0,27	−0,14	−0,15
Kamenec	0,52	0,55	–	−0,23	−0,23	−0,26	−0,14	−0,15

	Vrb	Lip	Kam	Wen	Lin	Esch	Odr	Bory
Wendelstedt	−0,27	−0,22	−0,23	–	0,56	0,70	−0,12	−0,17
Lindenau	−0,27	−0,23	−0,23	0,56	–	0,71	−0,12	−0,15
Eschfeld	−0,26	−0,27	−0,26	0,70	0,71	–	−0,15	−0,15
Odrzyca	−0,15	−0,14	−0,14	−0,12	−0,12	−0,15	–	0,48
Bory	−0,07	−0,15	−0,15	−0,17	−0,15	−0,15	0,48	–

`odhad`; `aia-sociomap-5`, the judgement's correlations, centred per future. Two of 28 pairs reliable. Split-half 0,91 against 0,59. Stability: 0,95 / 0,84 / 0,75 between the three settings, the last under 0,8: failed.

B.5 T8 · Shared failure of the two leading sites

When this site is worse than buying …	Vrb	Lip	Kam	Wen	Lin	Esch	Odr	Bory
Odrzyca	90 %	84 %	79 %	98 %	97 %	99 %	–	91 %
Kamenec	99 %	93 %	–	98 %	98 %	100 %	75 %	83 %

`odhad`; shares of the futures, uncentred. No site is a backup of another.

B.6 The decision boundary and the input table in brief

Compensation chance	Rule chooses	P(on time) Lipová / Kamenec / Odrzyca
0 %	Odrzyca	0,942 / 0,942 / 0,942
10 %	Odrzyca	0,852 / 0,851 / 0,844
14 %	Odrzyca	0,817 / 0,813 / 0,808
15 %	Kamenec	0,807 / 0,804 / 0,797
16 %	nothing	0,797 / 0,795 / 0,787
20 % (the model)	nothing	0,760 / 0,759 / 0,748

`odhad` (R2-P06 § 5.3). T9 holds 54 inputs: 16 public inputs await verification (*prozatímní*), 22 inputs the files do not hold are judgements (*úsudek*), the rest are read from client data (*ověřeno*); a public input can also be a judgement, so the groups overlap. Overrun and delay: median 17 %, 90th percentile 30,2 %, delay median 3 months, from 36 comparable plants, drawn together.

APPENDIX C

Board index

The boards of the Design canvas *AIA Obchodní model* after the sessions of 8 November, in the order of its rows, with the session that changed each. "Landed" means the session's branch is pushed and this document is aligned to it; "in progress" means the session is still working and the chapters that cite the board will be aligned when it lands. Boards marked parked stay on the canvas as planned, not deleted. Since 8 November a board that carries ChemVitro does so beside its Velis sections, each section marked with its case.

Code	Board	File	Session	State after 8 November
Z0	*Zadání: průchod*	`Z0Flow.dc.html`	A	landed: the flow and the board list around the spine
Z1	*Zadání: začátek*	`Z1Start.dc.html`	—	unchanged
Z2	*Zadání: triáž*	`Z2Triage.dc.html`	—	unchanged
Z3	*Zadání: rozhodnutí a pravidlo*	`Z3Decision.dc.html`	B	landed: § 03b the type-2 rule with four conditions as gates and the interpretation record
Z4	*Zadání: analýzy do zprávy*	`Z4Analyses.dc.html`	—	unchanged
Z5	*Zadání: obchodní model*	`Z5Model.dc.html`	B	landed: § 04b the parts of the cost-of-supply model, the timeline, the gate, the views, the perspectives
Z7	*Zadání: připravenost a předání*	`Z7Ready.dc.html`	—	unchanged
M1	*Model: Data*	`KM1Data.dc.html`	B	landed: § 01b–03b the document register, what the model takes with file and row, the 42 findings with dispositions
MV1	*Data: datová smlouva*	`MV1Contract.dc.html`	B	landed: § 01b–03b the contract and quality report of the purchases file, units and currencies as corrections
MV2	*Rešerše: typované vstupy*	`MV2PublicData.dc.html`	B	landed: § 01b the sixteen provisional inputs, § 02b verification at the source as a step
MV2b	*Rešerše: zjištění podle analýz*	`MV2bFindings.dc.html`	—	unchanged
M2	*Model: Předpoklady*	`KM2Assumptions.dc.html`	B	landed: § 01b T9 (67 rows, the K6 finding), § 02b uncertainty types, § 03b the estimate from comparables, § 03c the scope check, § 03d correlations and the stability test declared
M3	*Model: Model*	`KM3Model.dc.html`	C	landed: the shared timeline and the eleven items of the cost of supply
M4	*Model: Výsledky*	`KM4Results.dc.html`	C	landed: T1 at 8 %, T2 and T3, the decision boundary, delays, tornado, break-even, value of information
M4b	*Výsledky: Business case a finance*	`KM7Finance.dc.html`	C	landed: T4 and T5 with P(investment ≤ 175)
M4c	*Výsledky: Účetnictví*	`KM8Accounting.dc.html`	C	parked for type 2
M4d	*Výsledky: Operační výzkum*	`KM9Operations.dc.html`	C	landed: T6 with each column named as its component
M4g	*Výsledky: Harmonogram*	`KM12Schedule.dc.html`	C	new, landed: the start against the deadline, what lengthens each tail, delay × hiring as one grid
M4f	*Výsledky: Logistika*	`KM11Logistics.dc.html`	C	landed: product freight per site, the Odrzyca spur, the Eschfeld bridge, the trucking surcharge, the logistics rows of T9
M4e	*Výsledky: Velikost trhu*	`KM10Market.dc.html`	C	parked for type 2
MV3	*Výsledky: Sociomapa systémů*	`MV3SystemMap.dc.html`	D	in progress: the futures map as the exemplar draws it, T7, T8, the stability test
MV9	*Mapa scénářů vah*	new	D	in progress: the weighting-scenario map on the real engine

Code	Board	File	Session	State after 8 November
SI1–SI4	*Sociomapa z indikátorů*: method, matrix, options map, suppliers map	`SI1Method` … `SI4Suppliers.dc.html`	D	in progress: reduced to what MV9 reads
MV4	*Osnova zprávy*	`MV4XRay.dc.html`	E	landed: the 18 parts R2-P01…P18 with typed records, the grades with their attributes, the argument's five parts mapped; one alignment commit pending
MV8	*Zpráva: kontrola platnosti modelu*	`MV8Validity.dc.html`	E	landed: the ten limits of chapter 14 (3 caught, 7 open), the five questions, seven extreme-input computations, K1–K6
M6	*Model: Zpráva*	`KM6Report.dc.html`	E	landed: report 2 as rendered, the release state with four qualifications, today against new in the renderer
MV0	*Definované MVP: přehled*	`MV0Scope.dc.html`	A	landed: rewritten around the spine; acceptance = the answer key
MV5	*Metodika: validační sada a otázky*	`MV5Validation.dc.html`	E	landed: the register entries report 2 waits on, six questions, generators G7–G10, the release rule
MV6	*Katalog analýz*	`MV6Catalogue.dc.html`	—	unchanged; the catalogue beyond the perspectives is parked

Sizes and positions: `canvas.json` on each session's branch; the coordinator merges them. Reference renders of version 51: `reference/`. The canvas is published by the coordinator from the exported files; no session publishes it.